Some open problem in cosmic-ray physics

(Lecture 10)

Satyendra Thoudam
Khalifa University

Outline of this chapter

- Introduction: Standard model of cosmic rays.
- Surprises from recent cosmic-ray measurements

Introduction: Standard model of cosmic rays in the Galaxy

CR transport equation in the Galaxy

$$-\nabla.(D\nabla N) + \frac{N}{T_{\rm nn}} - \frac{\partial}{\partial E}\left[b(E)N\right] = Q$$

Diffusion (escape)	CR-ISM interaction	Energy loss (e-)	Source
$D \propto E^{a}$		$b(E) \propto E^{2}$	$E^{-\gamma}$
$T_{\rm esc} \sim \frac{H^2}{D} \propto E^{-a}$		$T_{\rm loss} \propto \frac{1}{E}$	

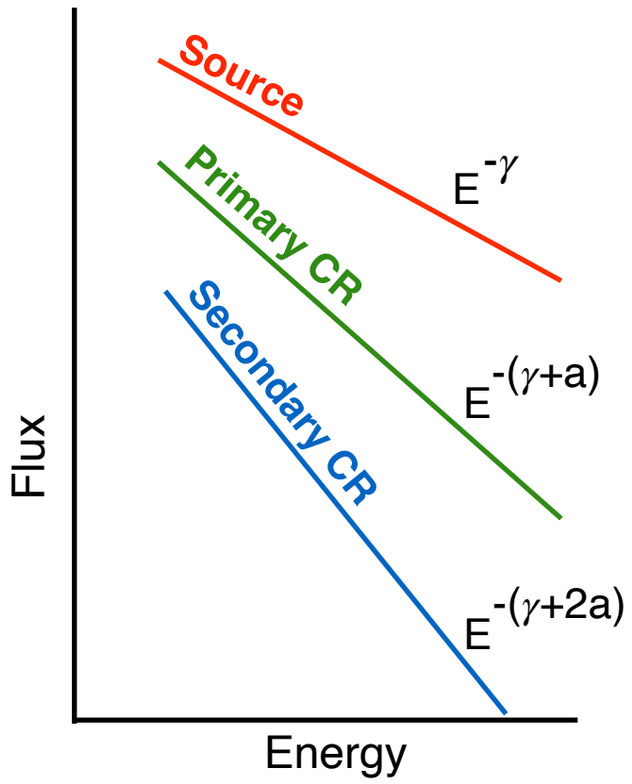
Spectrum after propagation

For CR nuclei:	✓	✗	✗	→	$\frac{N}{T_{\rm esc}} = Q \Rightarrow N \propto E^{-(\gamma+a)}$
For electrons:	✗	✗	✓	→	$\frac{N}{T_{\rm loss}} = Q \Rightarrow N \propto E^{-(\gamma+1)}$

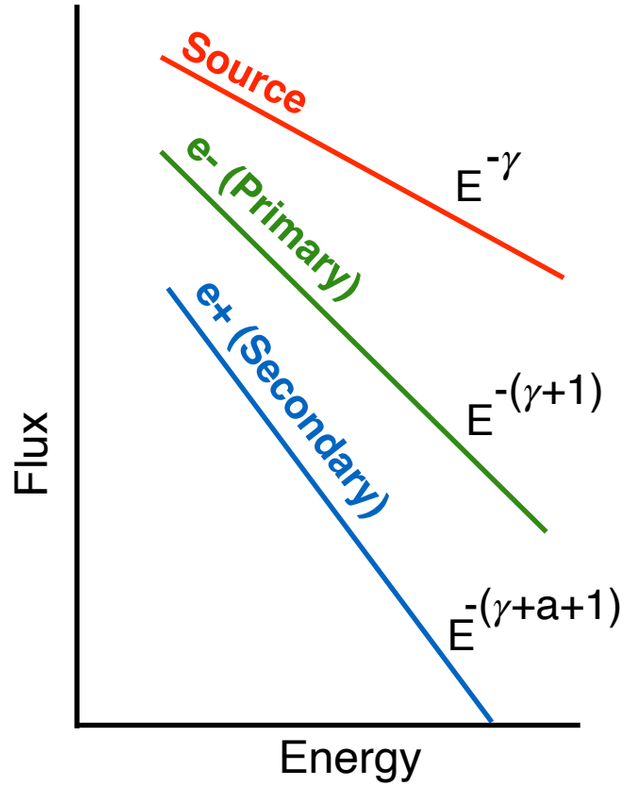
Power law

Introduction: Standard model of cosmic rays in the Galaxy

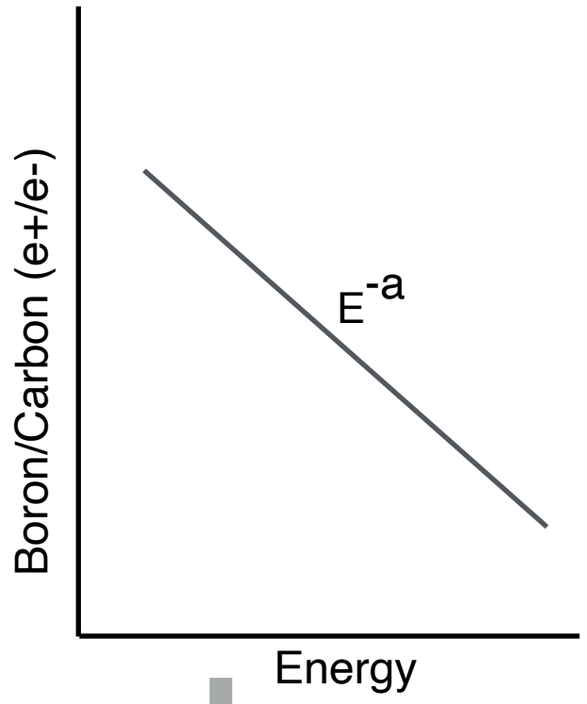
CR nuclei



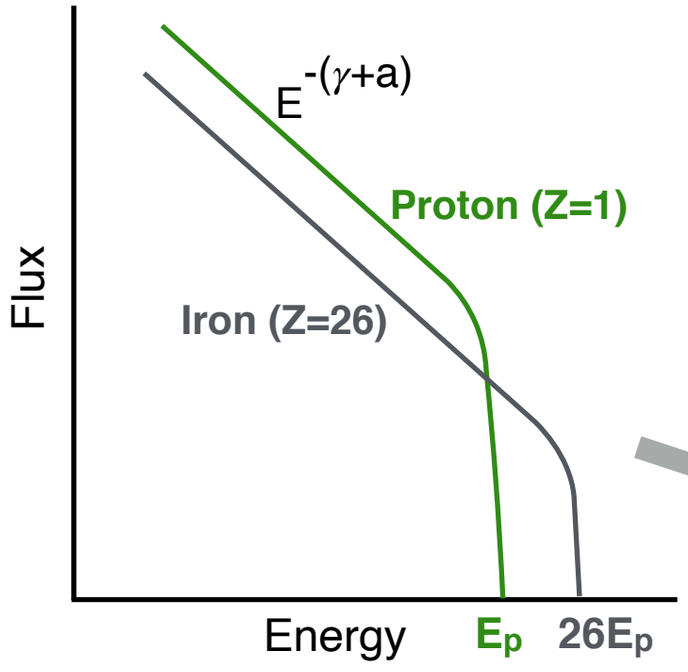
CR electrons



Secondary/Primary ratio

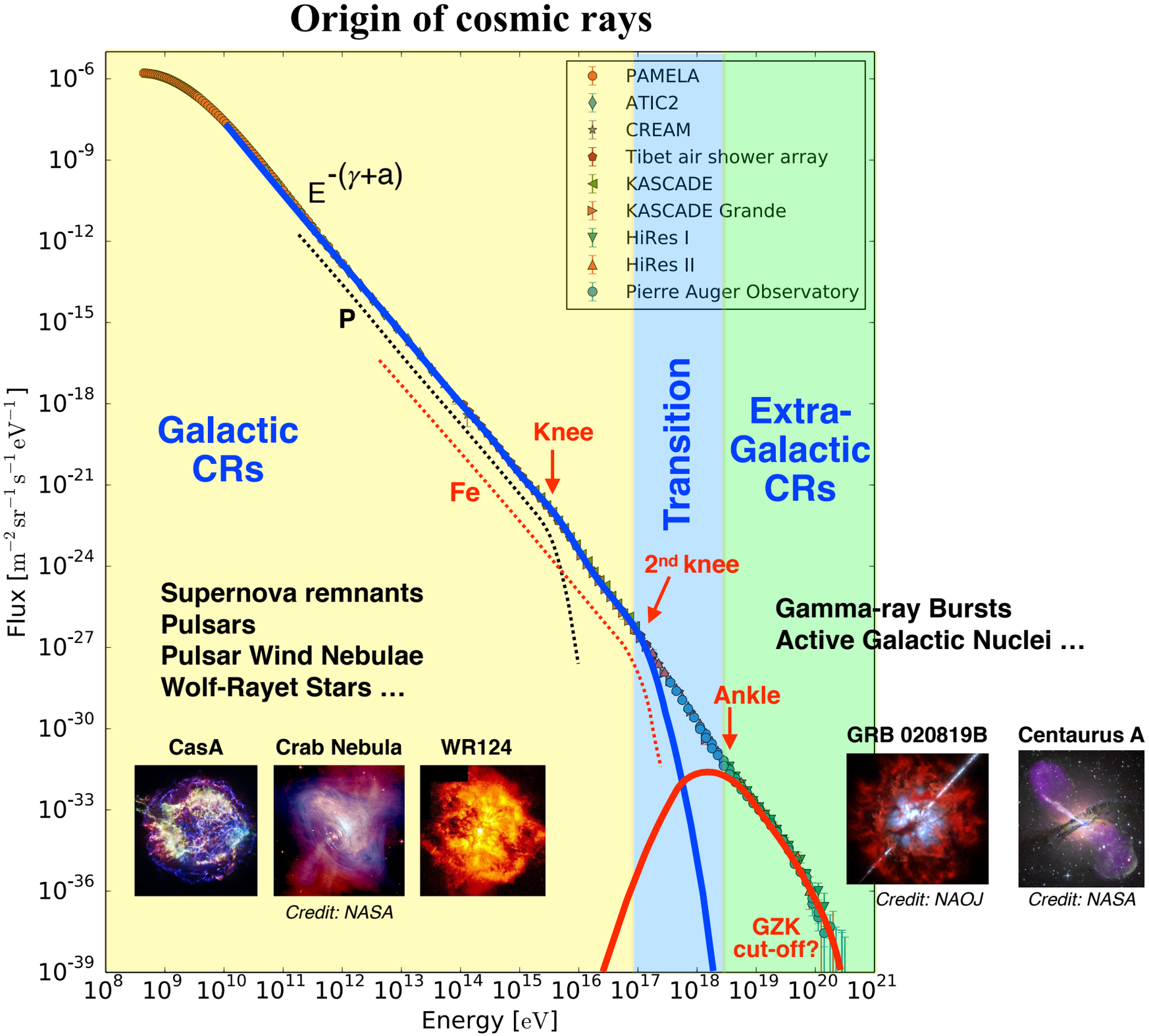


- ✦ Independent of source spectrum.
- ✦ Measure of the diffusion coefficient => CR escape time from the Galaxy.
- ✦ Measurement of B/C gives: a≈ 0.33



Composition should become heavier as energy increases

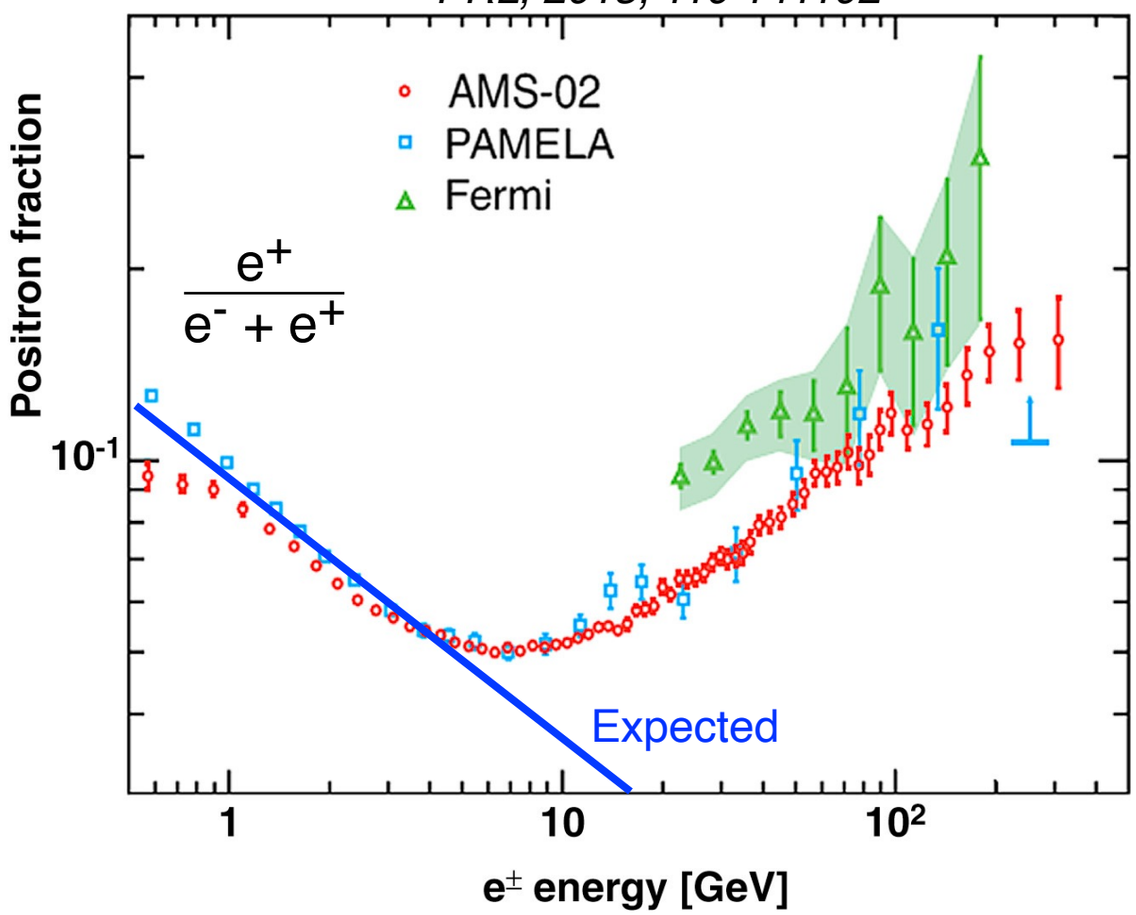
Introduction: Standard model of cosmic rays in the Galaxy



Some surprises from recent measurements

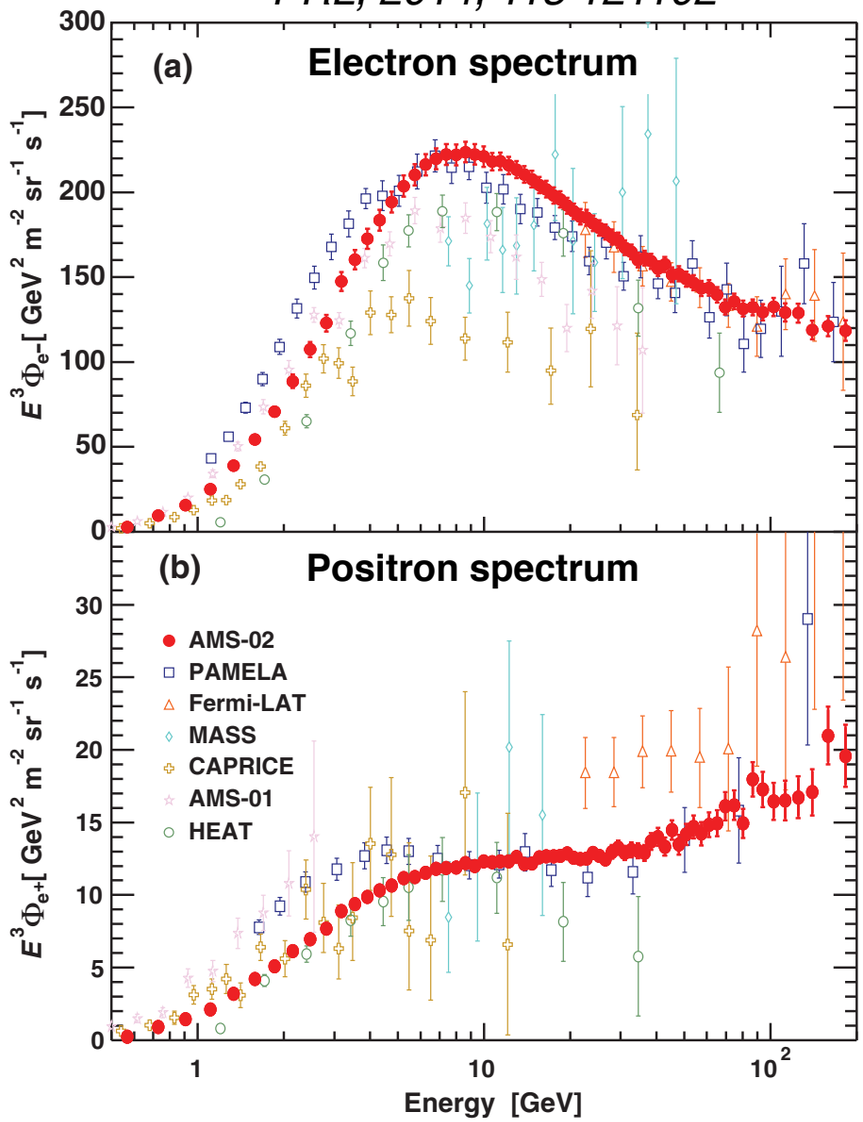
(I) Rise in the positron fraction above ~ 10 GeV

AMS-02 Collaboration
PRL, 2013, 110 141102



- ✦ Observed by PAMELA, AMS-02, Fermi, ..
- ✦ Continuous rise above ~ 10 GeV
- ✦ Theoretically expected to decrease as ~ $E^{-0.33}$
- ✦ Rise due to increase in the e+ flux.

AMS-02 Collaboration
PRL, 2014, 113 121102

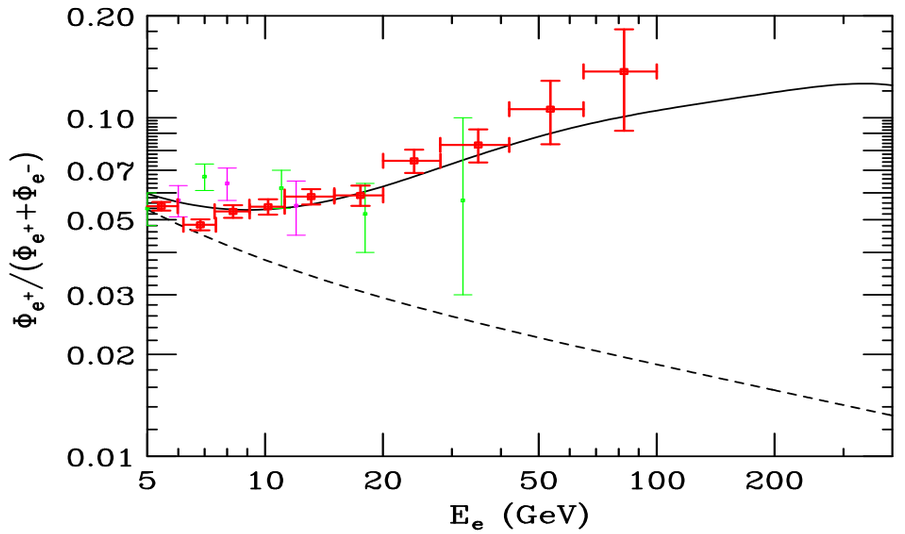


(I) Rise in the positron fraction above ~ 10 GeV

Some possible explanations

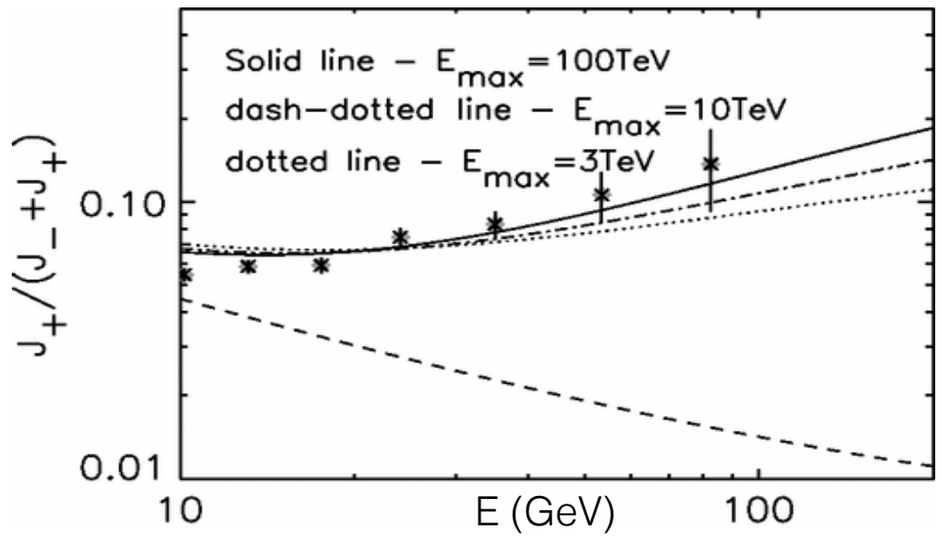
✦ Contribution from nearby pulsars:

Hooper et al. 2009, JCAP, 01, 025
Linden & Profumo, 2013, ApJ,772, 18



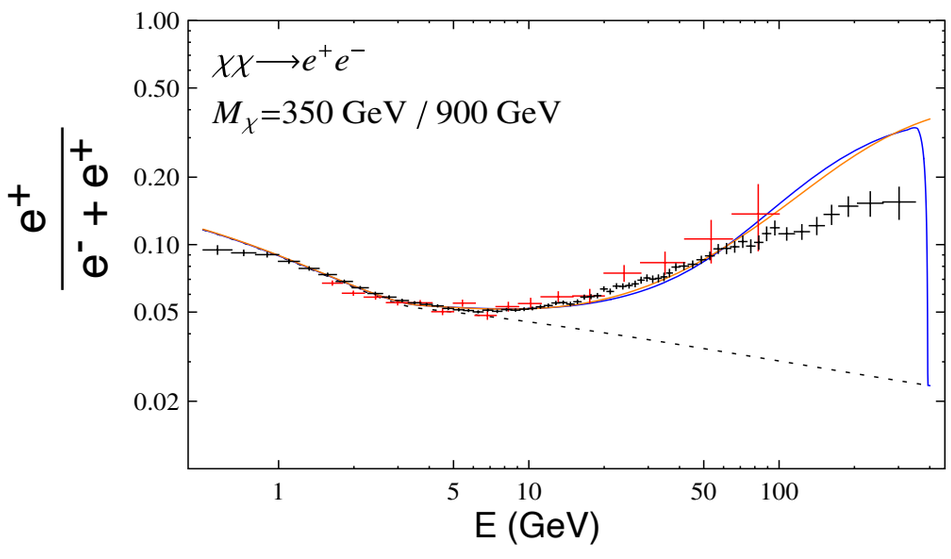
✦ Secondary e^+ production inside sources:

Blasi 2009, PRL, 103, 051104

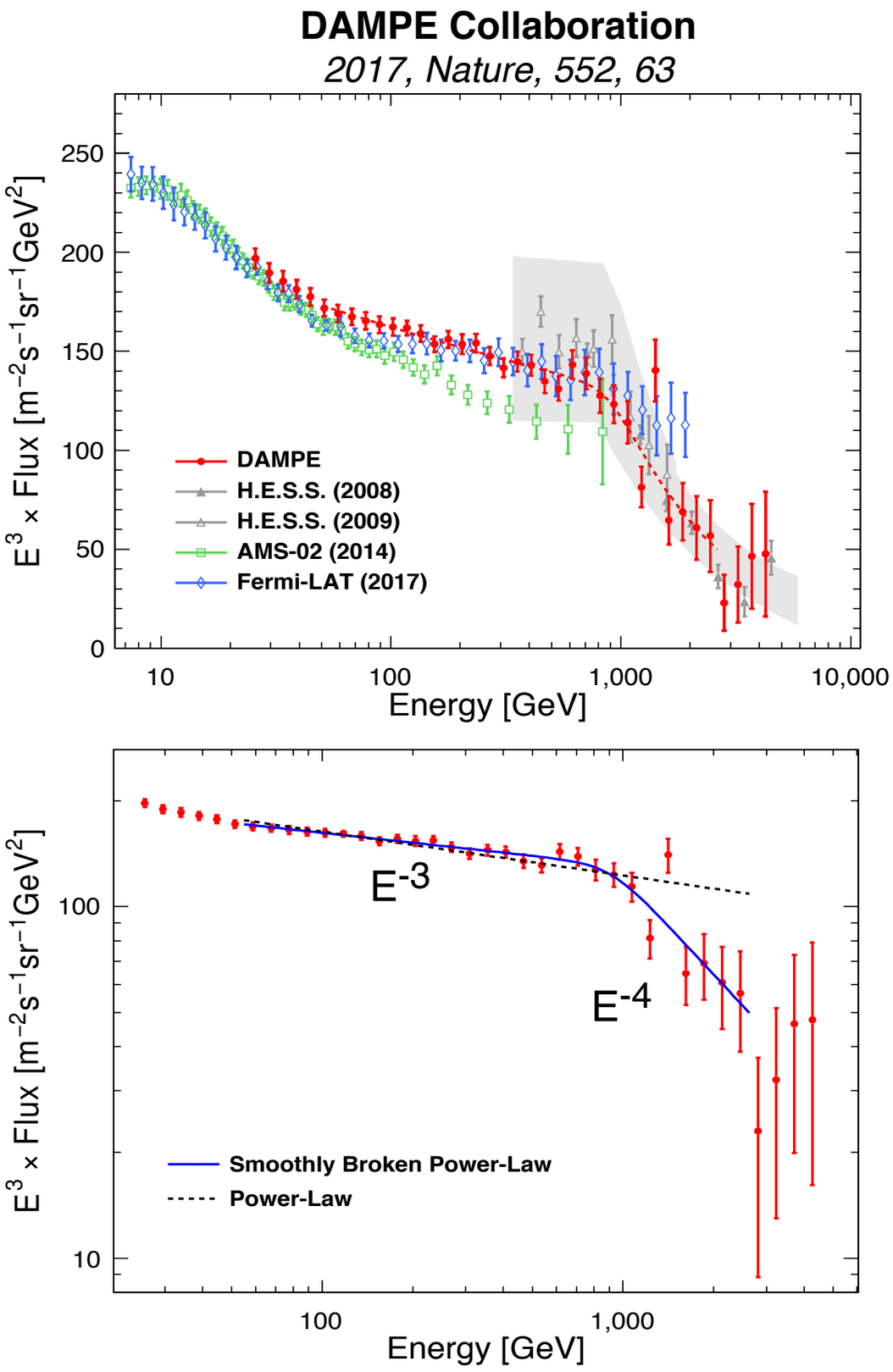


✦ Dark matter annihilation:

Cholis & Hooper 2013, PRL, 88, 023013

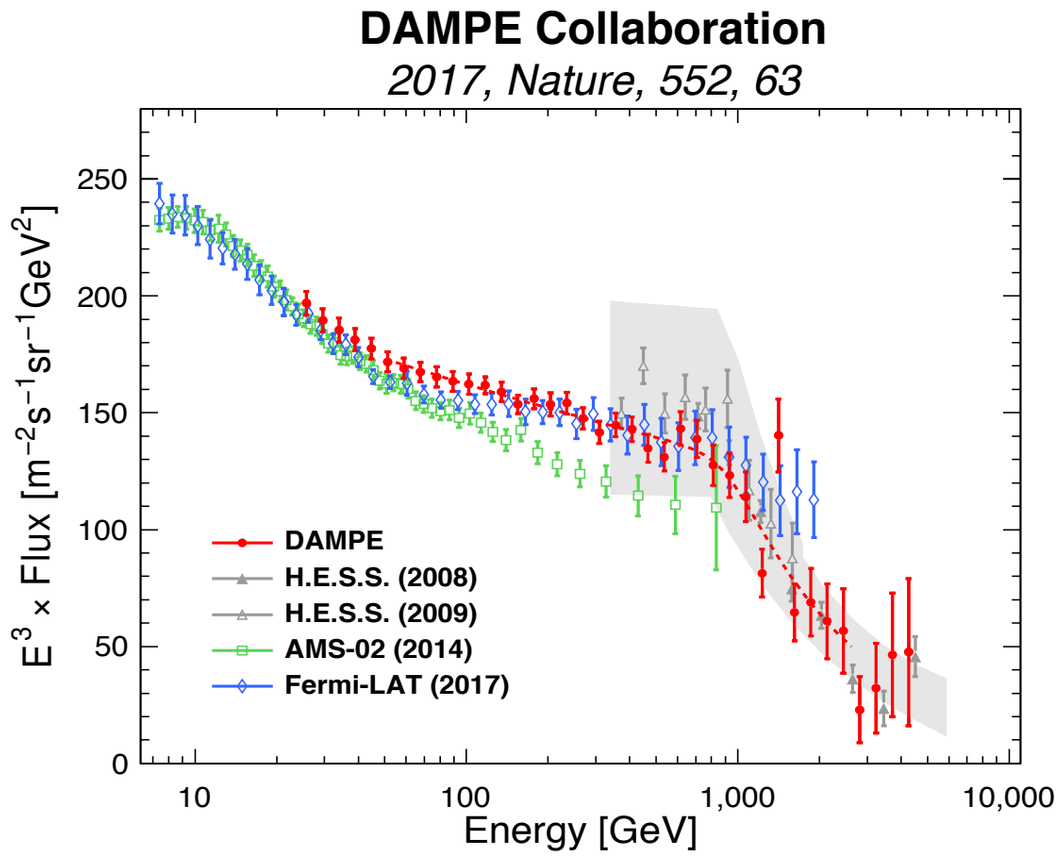


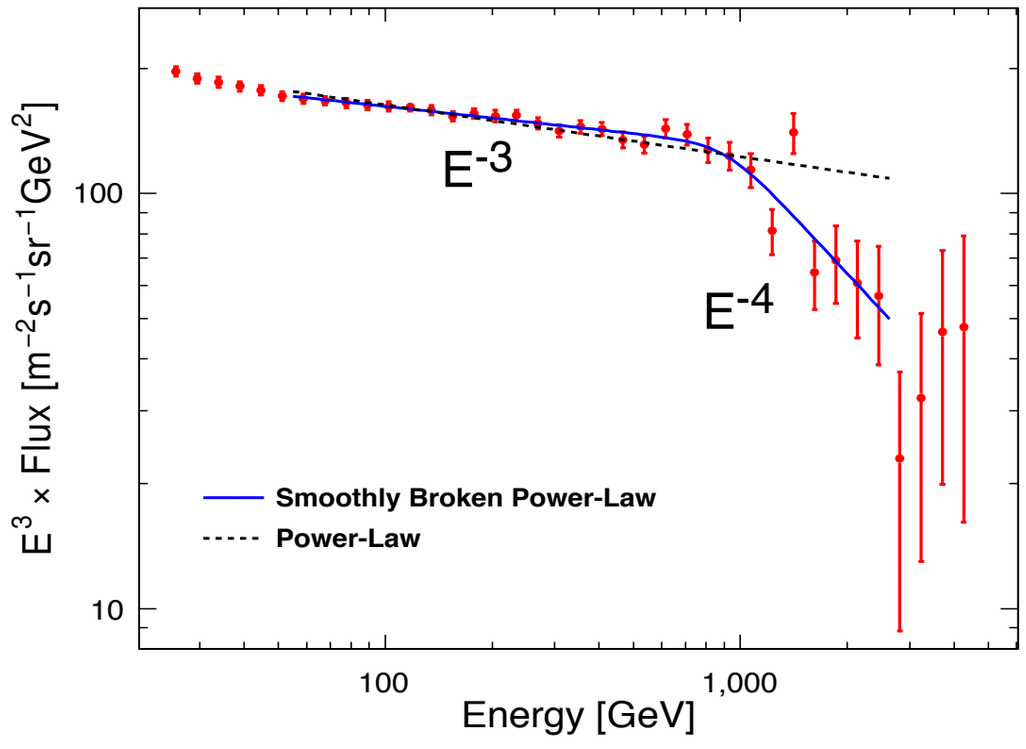
(II) Break in the all-electron spectrum at ~ 1 TeV



- ✦ Observed by H.E.S.S., CALET and DAMPE.
- ✦ Spectral index:
  - $\Gamma \approx -3$, below break.
  - $\Gamma \approx -4$, above break.
- ✦ Unlikely to be propagation effect in the Galaxy.
- ✦ Unlikely due to cut-off in the positron flux.

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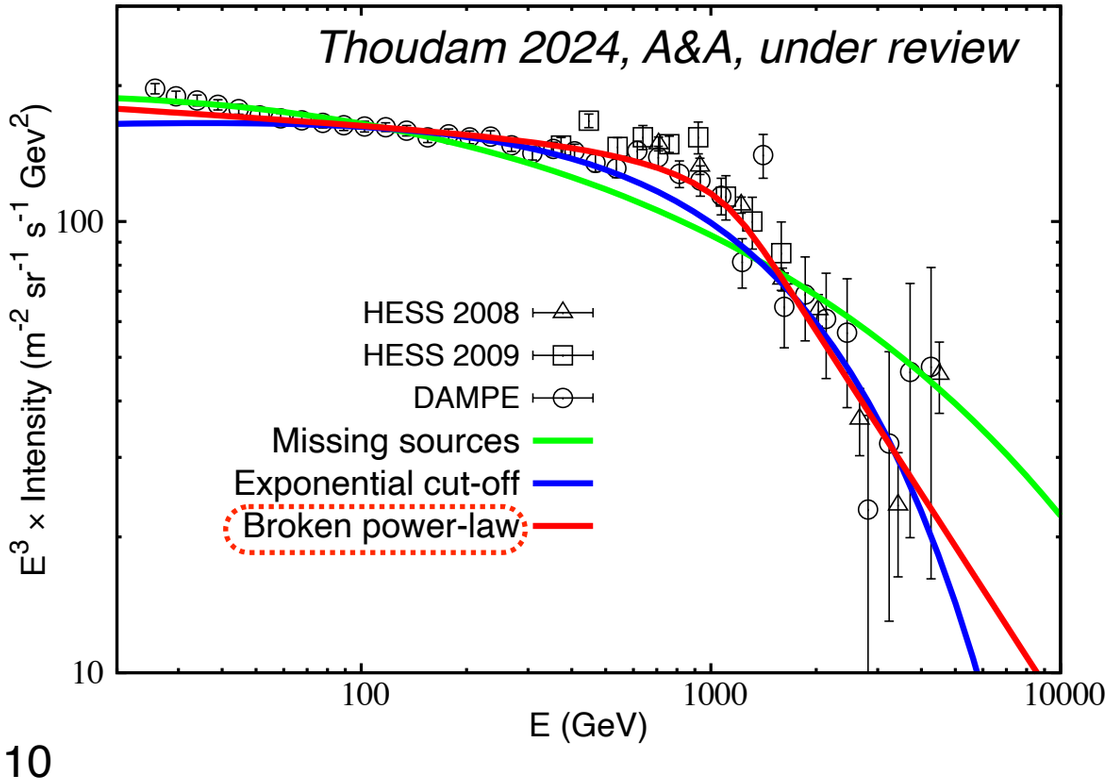




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Possible astrophysical origin

- ✦ Missing nearby sources
- ✦ Exponential cut-off in the source spectrum
- ✦ **Broken power-law source spectrum** ✓



(II) Break in the all-electron spectrum at ~ 1 TeV

… but, how do we get a broken power-law source spectrum ?

Thoudam & Hörandel 2011, MNRAS, 414, 1432

A toy model for particle spectrum inside the sources

- ✦ CRs are accelerated by supernova shock waves.
- ✦ CRs escape downstream of the shock & remain confined.
- ✦ High-energy electrons suffer radiative losses.
- ✦ CRs are released when the shock slows down.

$$\frac{\partial N_e}{\partial t} = \frac{\partial}{\partial E}\{b(E)N_e\} + q_e,$$

Energy loss $b(E) = kE^2$ — Accel^n $\propto E^{-\Gamma}$

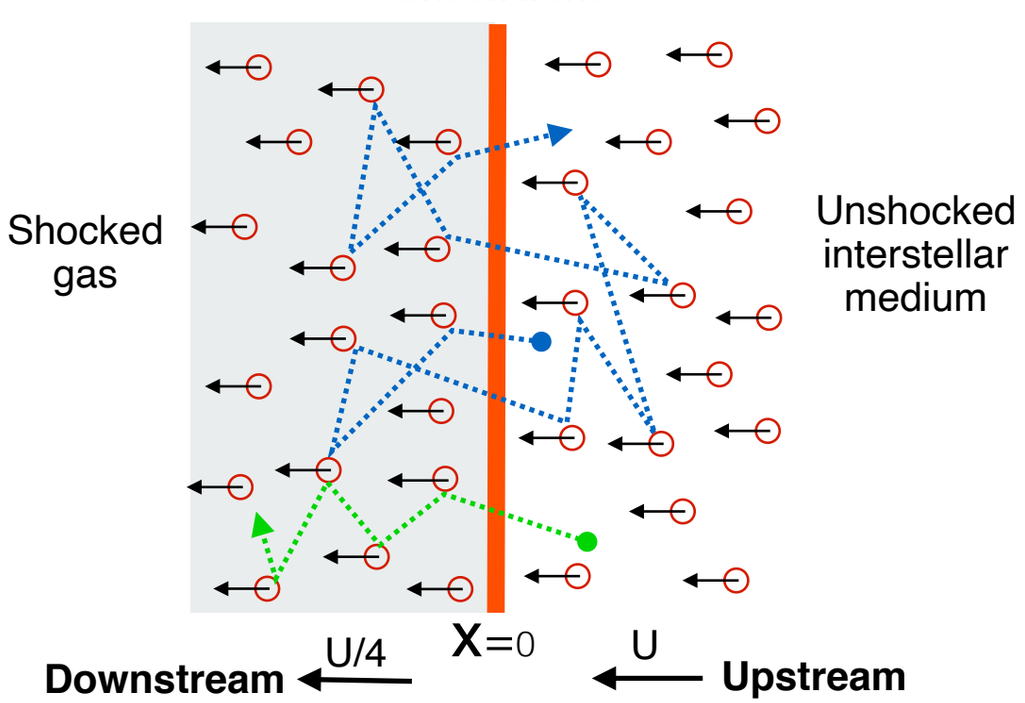
Spectral evolution in the downstream region

$$N_e(E,t) \propto E^{-\Gamma} \quad E < 1/kt$$

$$N_e(E,t) \propto E^{-(\Gamma+1)} \quad E \geq 1/kt$$

Final source spectrum released in the Galaxy

Shock rest frame



(II) Break in the all-electron spectrum at ~ 1 TeV

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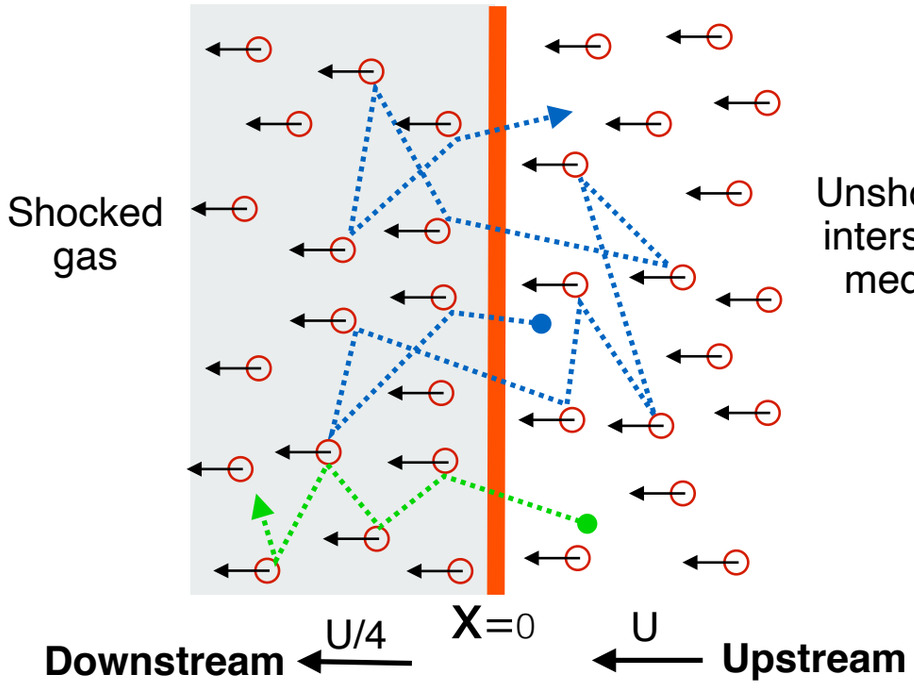
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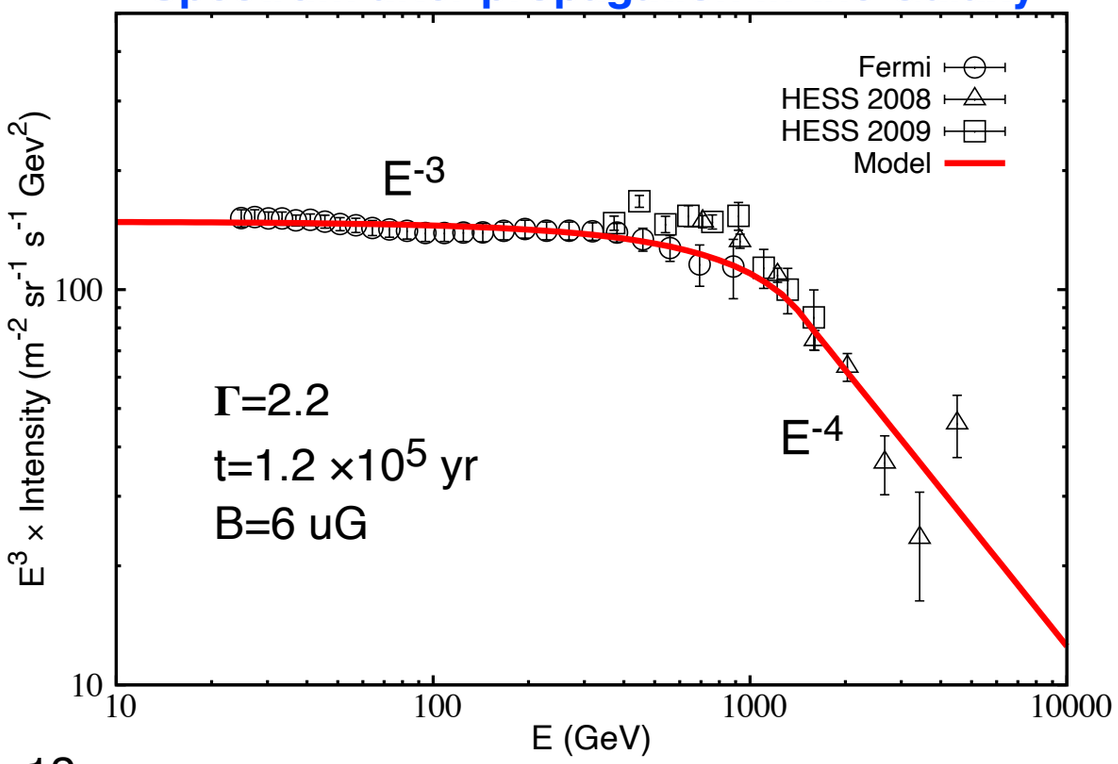
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Final source spectrum released in the Galaxy

Shock rest frame



Spectrum after propagation in the Galaxy



(II) Break in the all-electron spectrum at ~ 1 TeV

… but, how do we get a broken power-law source spectrum ?

A more realistic model

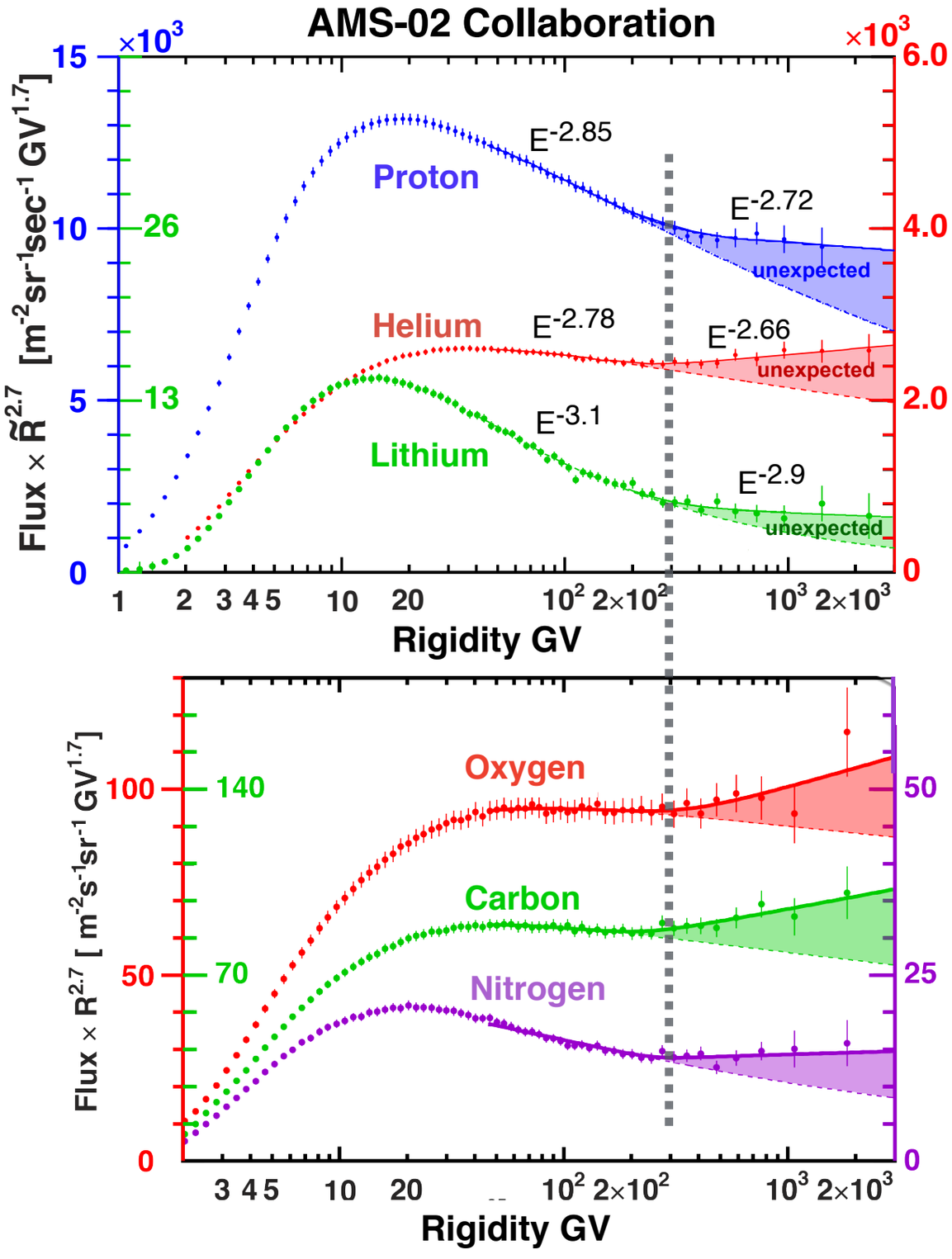
- ✦ Similar idea as in *Thoudam & Hörandel 2011*.
- ✦ But, considering diffusive shock acceleration + energy losses.
- ✦ Analytical approach:
  - No exact analytical solution exists so far.

Time-dependent DSA process including radiative energy losses

$$\frac{\partial}{\partial x}\left(D\frac{\partial f}{\partial x}\right) - u\frac{\partial f}{\partial x} + \frac{1}{3}\frac{du}{dx}p\frac{\partial f}{\partial p} + \frac{1}{p^2}\frac{\partial}{\partial p}\left[p^2 b(p) f\right] + Q(x,p,t) = \frac{\partial f}{\partial t}$$

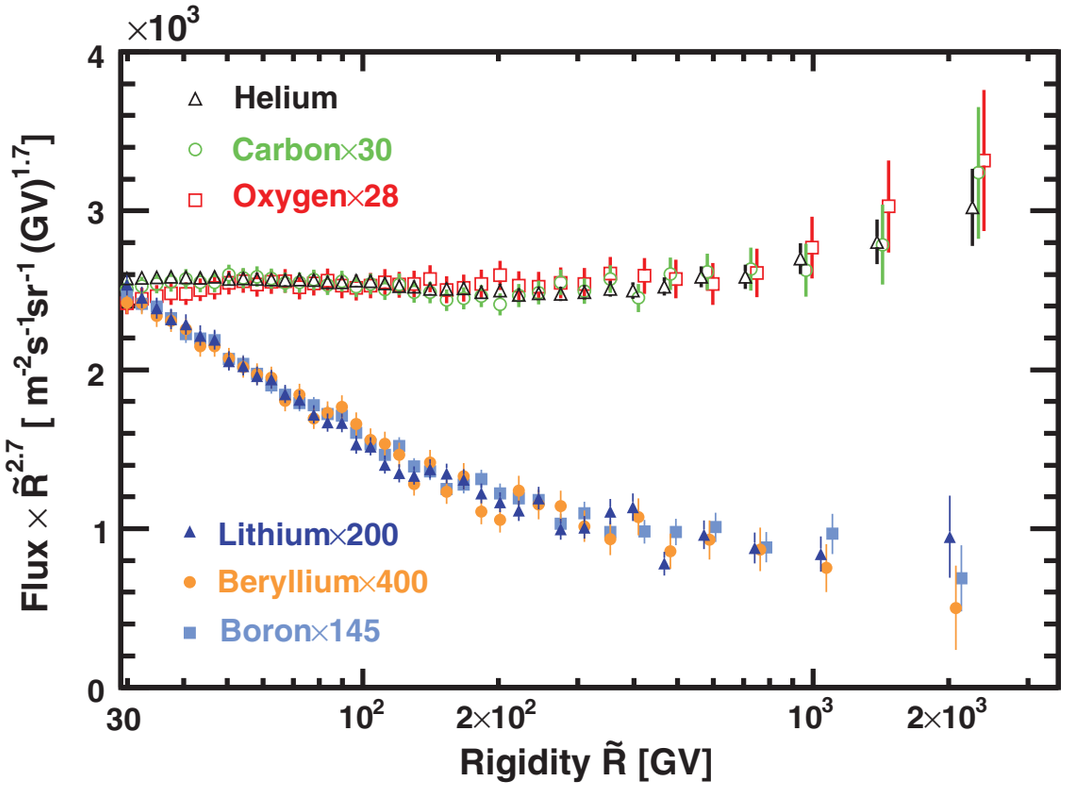
Energy dependent Diffusion	Advection	Compression	Energy loss	Injection

(III) Spectral hardening of CR nuclei at ~200 GeV/nucleon



- Hardening observed by PAMELA, CREAM, AMS-02.
- Observed in both primary and secondary nuclei.
- Change in the spectral index, $\triangle\Gamma\approx0.12$
- Proton spectrum steeper than He, C, O by $\triangle\Gamma\approx0.07$

2015, PRL, 114,171103
2015, PRL, 115, 211101
2017, PRL, 119, 251101
2018, PRL, 120, 021101



(III) Spectral hardening of CR nuclei at ~200 GeV/nucleon

Possible explanations

✦ **Effect of nearby sources:**

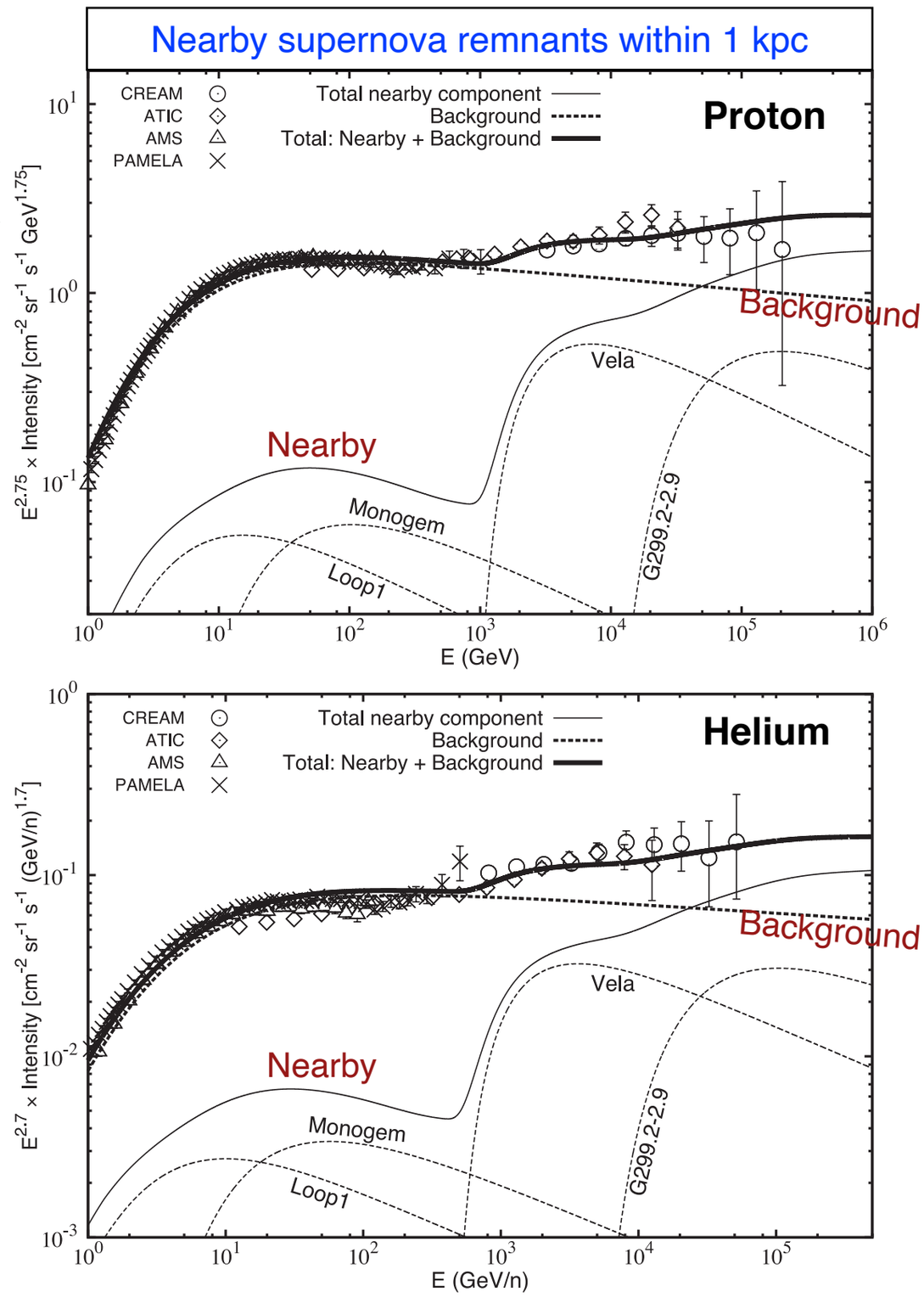
Thoudam & Hörandel 2012, MNRAS, 421, 1209
Thoudam & Hörandel 2013, MNRAS, 435, 2532

Model

CR observed=Background + Nearby component

SNRs within 1 kpc from the Earth

Name	Distance (kpc)	Age (yr)
Geminga	0.15	3.4×10^5
Loop1	0.17	2.0×10^5
Vela	0.30	1.1×10^4
Monogem	0.30	8.6×10^4
G299.2-2.9	0.50	5.0×10^3
Cygnus Loop	0.54	1.0×10^4
G114.3+0.3	0.70	4.1×10^4
Vela Junior	0.75	3.5×10^3
S147	0.80	4.6×10^3
HB9	0.80	6.6×10^3
HB21	0.80	1.9×10^4
SN185	0.95	1.8×10^3



(III) Spectral hardening of CR nuclei at ~200 GeV/nucleon

Possible explanations

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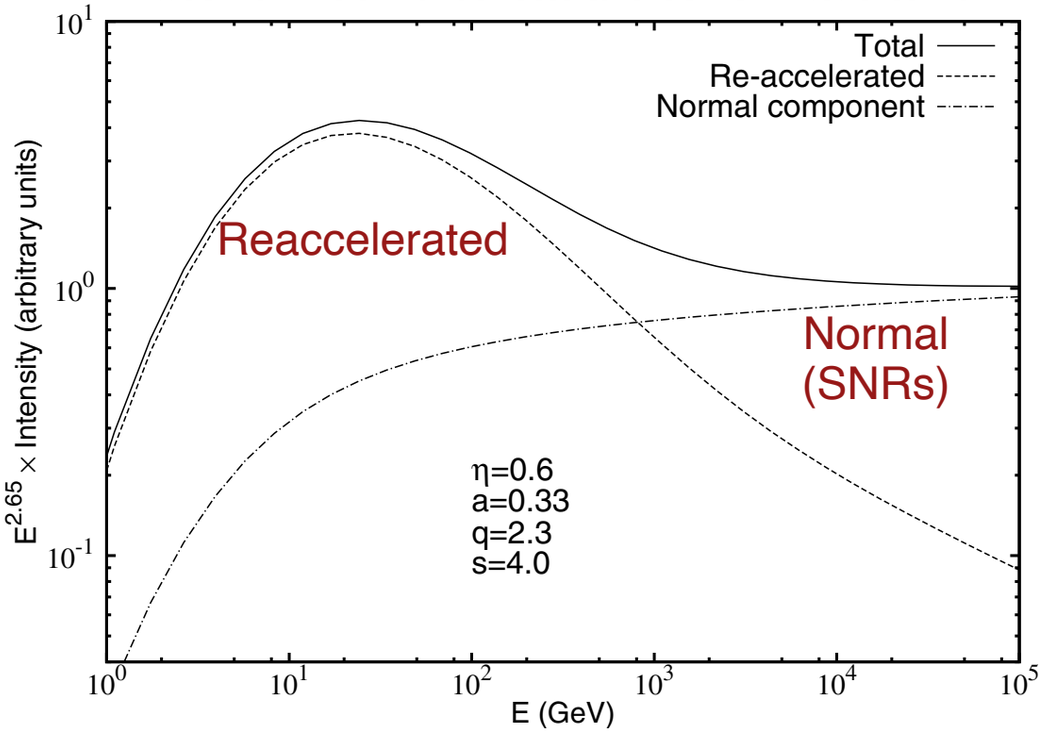
Thoudam & Hörandel 2012, MNRAS, 421, 1209
Thoudam & Hörandel 2013, MNRAS, 435, 2532

✦ **CR reacceleration by weak shocks in the Galaxy:**

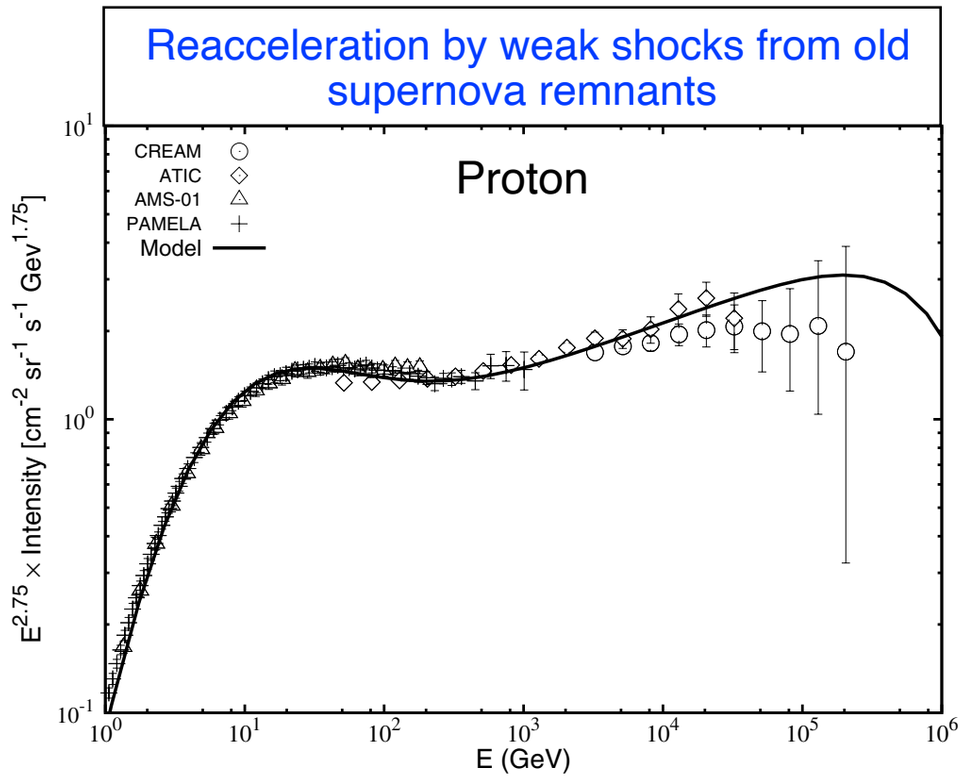
Thoudam & Hörandel 2014, A&A, 567, A33

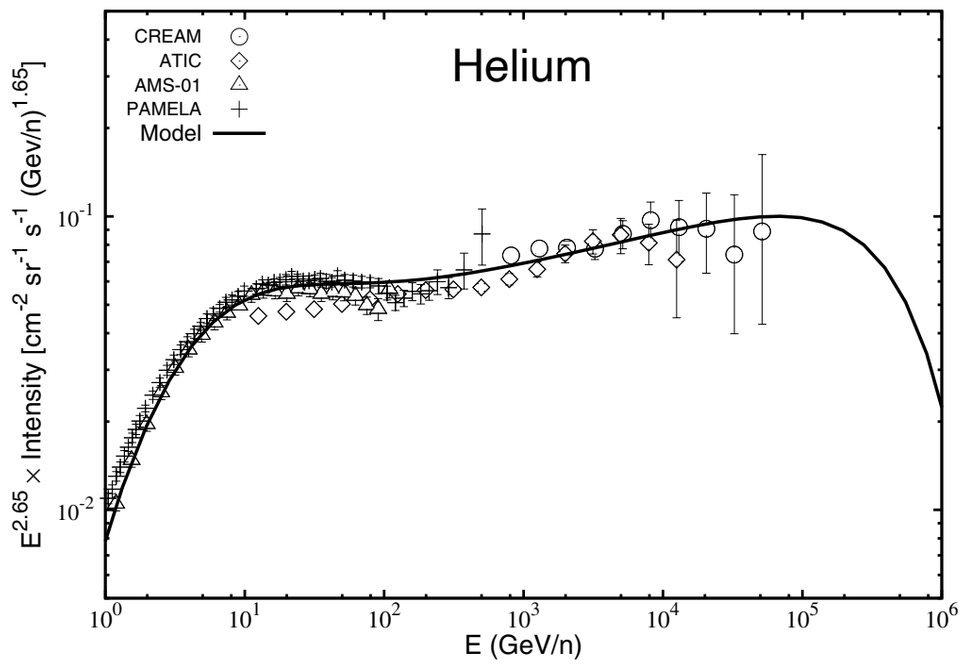
Model

CR observed=Normal + Reaccelerated



Reacceleration by weak shocks from old supernova remnants





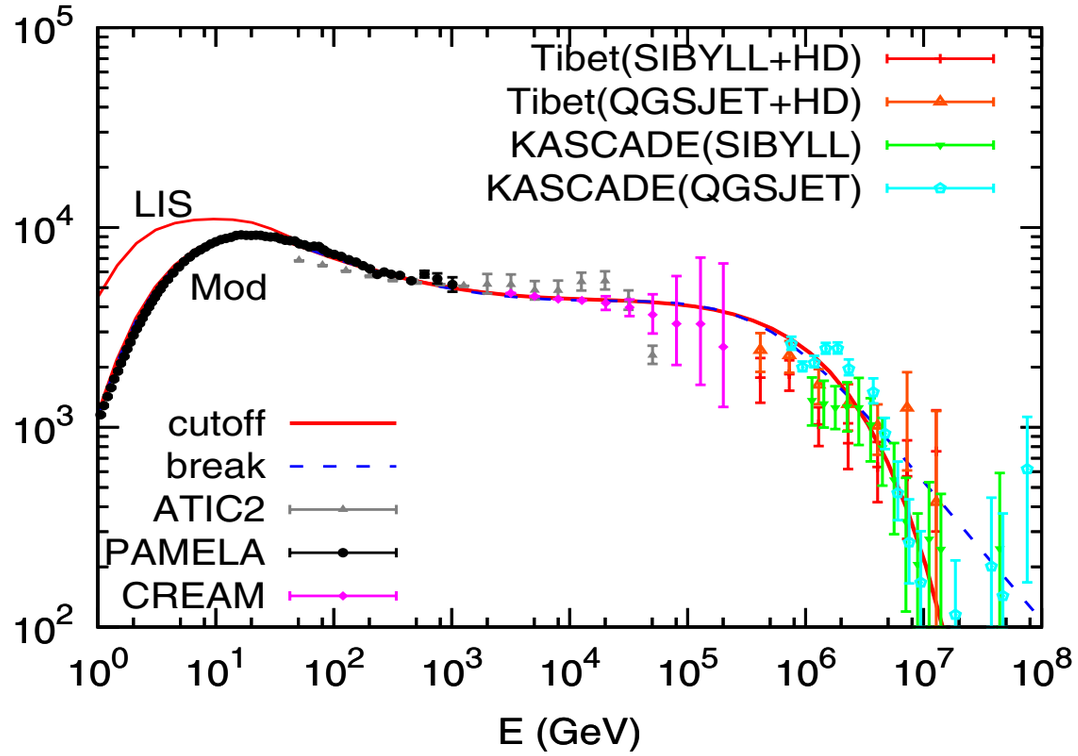
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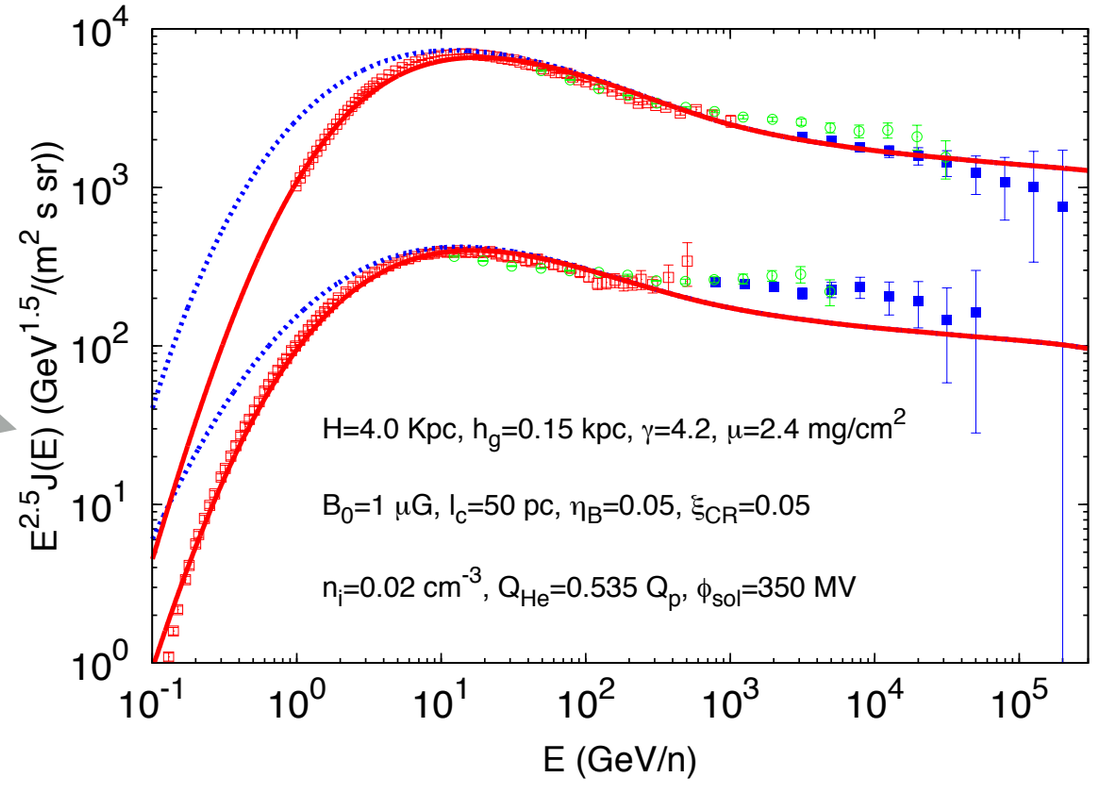
Possible explanations

- **Effect of nearby sources:**
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  Thoudam & Hörandel 2013, MNRAS, 435, 2532

- **CR reacceleration by weak shocks in the Galaxy:**
  Thoudam & Hörandel 2014, A&A, 567, A33

- **Break in the source spectrum:**
  Yuan et al. 2011, PRD, 84, 043002
  Ptuskin et al. 2013, ApJ, 763, 47
- **Break in the diffusion coefficient:**
  Tomassetti 2012, ApJL, 752, L13
  Blasi 2012, PRL, 109, 061101

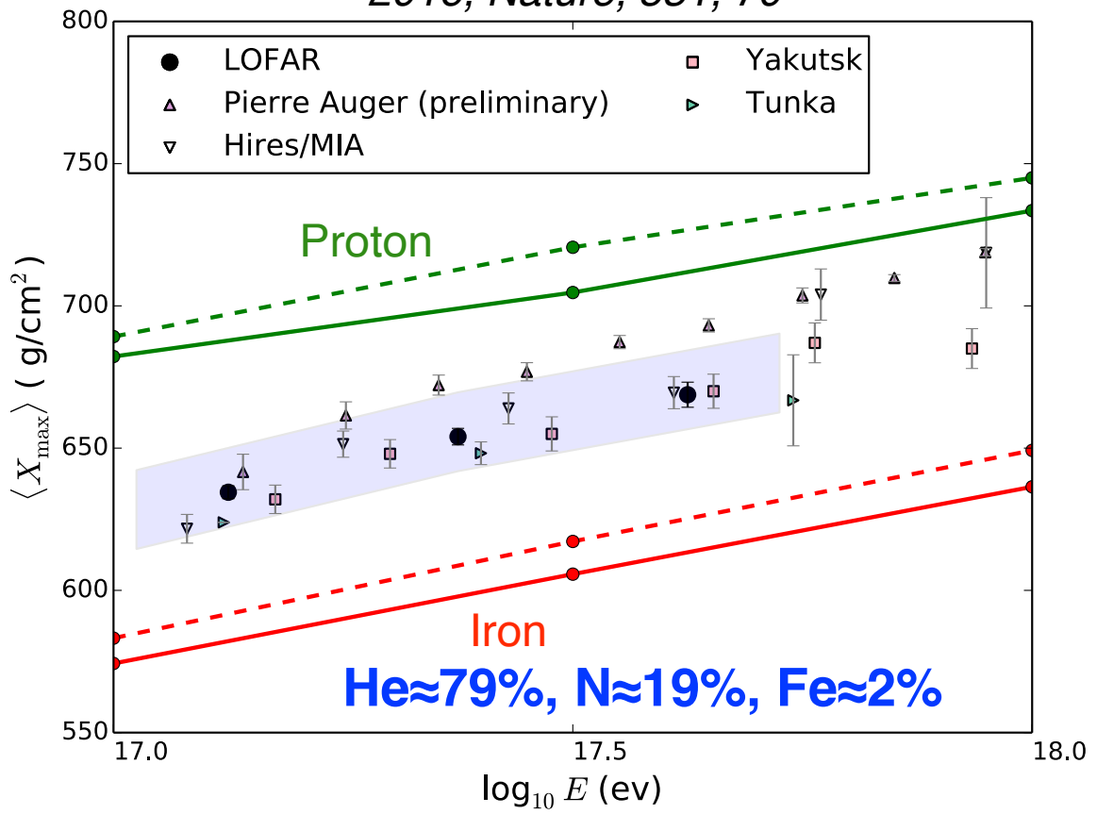




(IV) Dominant helium fraction in cosmic rays at ~ $10^{17\text{-}17.5}$ eV

LOFAR Collaboration

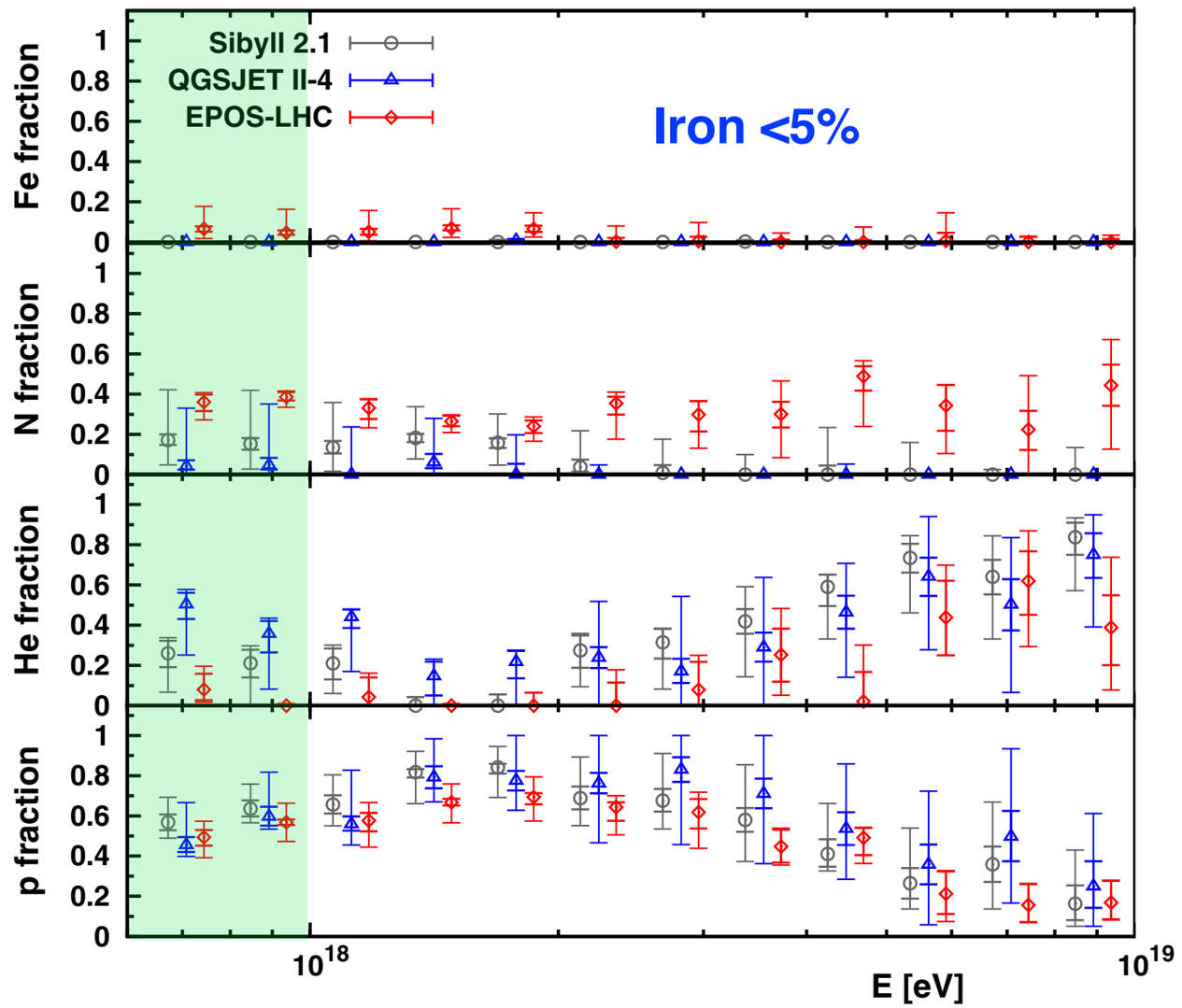
2016, Nature, 531, 70



- ✦ Theoretically expected to be iron dominant.
- ✦ Also agrees with the Pierre Auger measurements.

Pierre Auger Collaboration

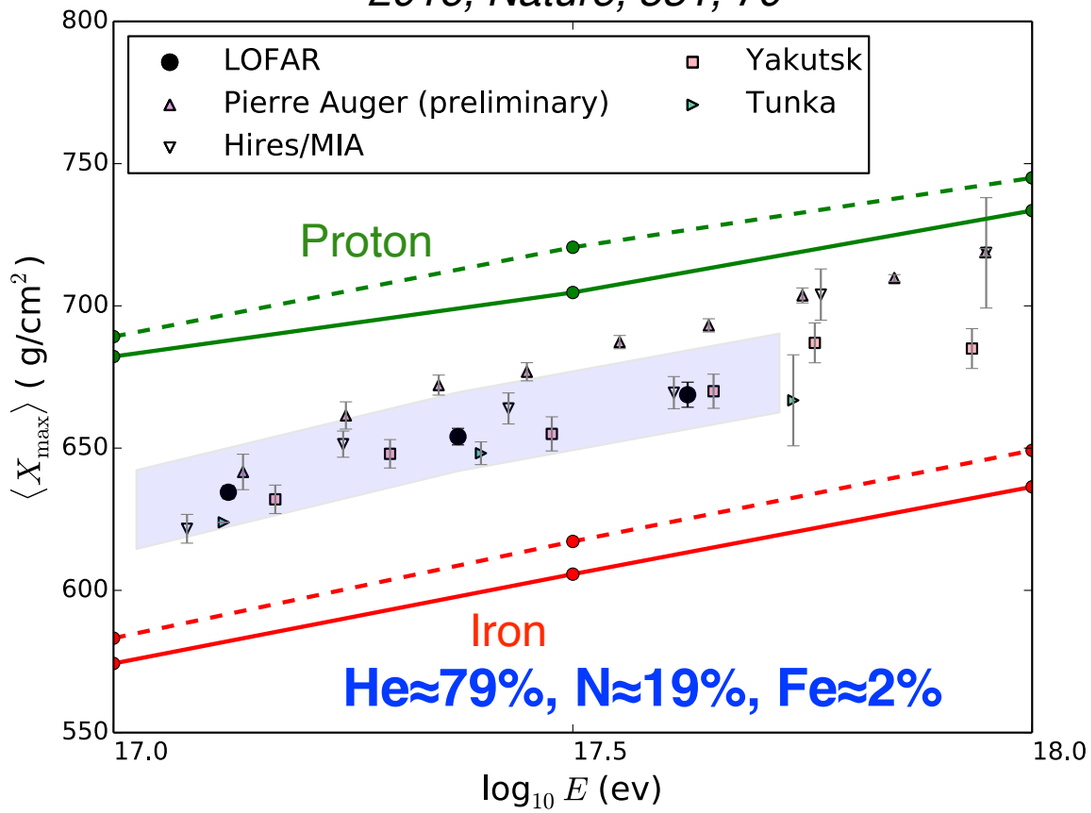
2014, PRD, 90, 122006



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LOFAR Collaboration

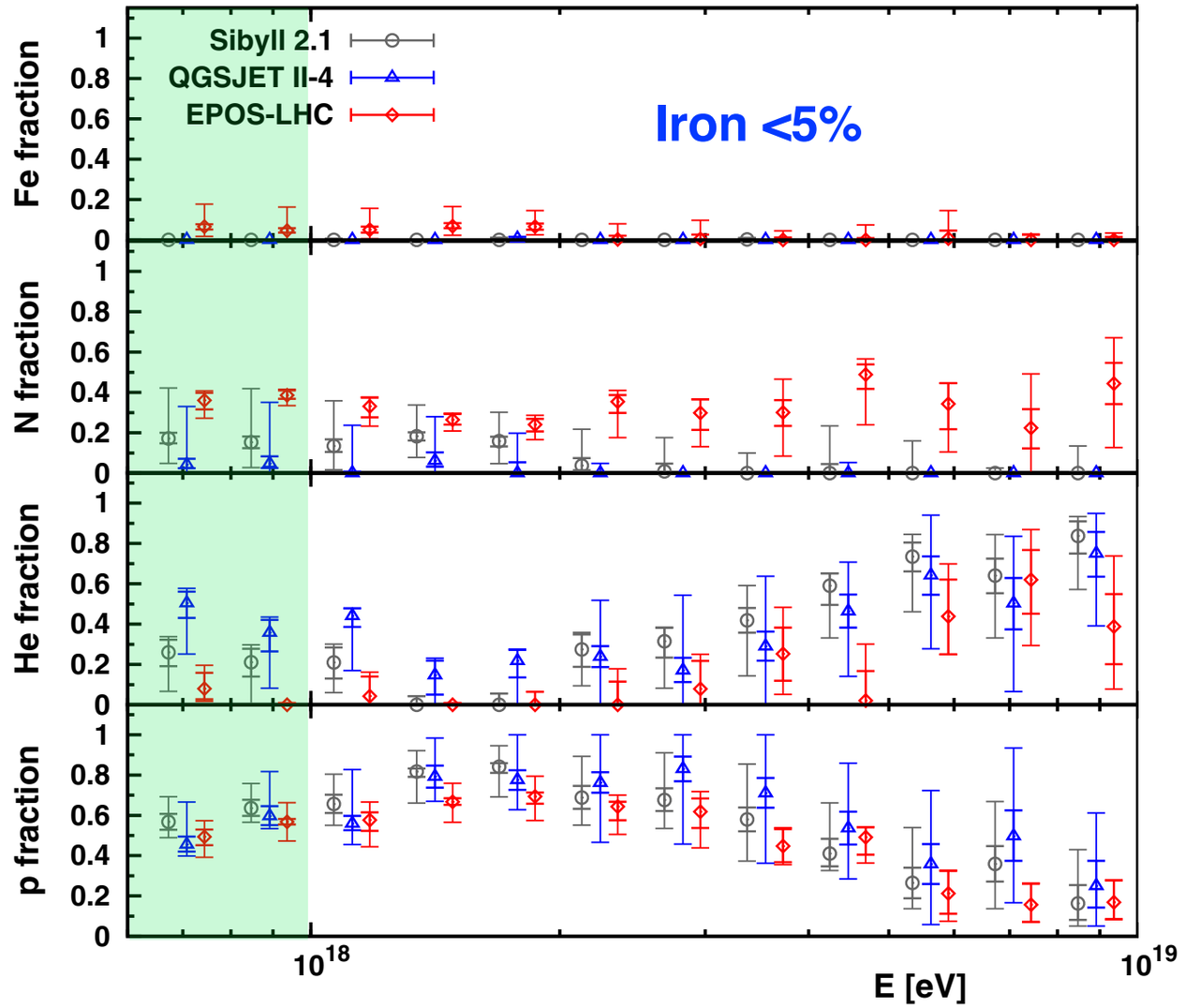
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Pierre Auger Collaboration

2014, PRD, 90, 122006



Possible explanations

- ✦ Multiple Galactic components? ✓
- ✦ Extra-galactic component? ✗

Thoudam+ 2016, A&A, 595, A33

(IV) Dominant helium fraction in cosmic rays at ~ $10^{17\text{-}17.5}$ eV

Possible explanation: A two-component model of Galactic cosmic rays

1st component: CRs from regular supernovae

- Supernova shocks in ISM.
- Diffusive shock acceleration.
- Composition <=> solar abundance.
- Typical frequency ~ 1/30 yr.
- Dominate at low energies.

SNR-CRs

2nd component: CRs from Wolf-Rayet (WR) supernovae

- Supernova shocks in WR star wind.
- Flatter spectrum (Eichmann & Rachen, ICRC 2021)
- Composition: Low protons.
- Typical frequency ~ 1/210 yr.
- Their contribution of CRs increase with energy.

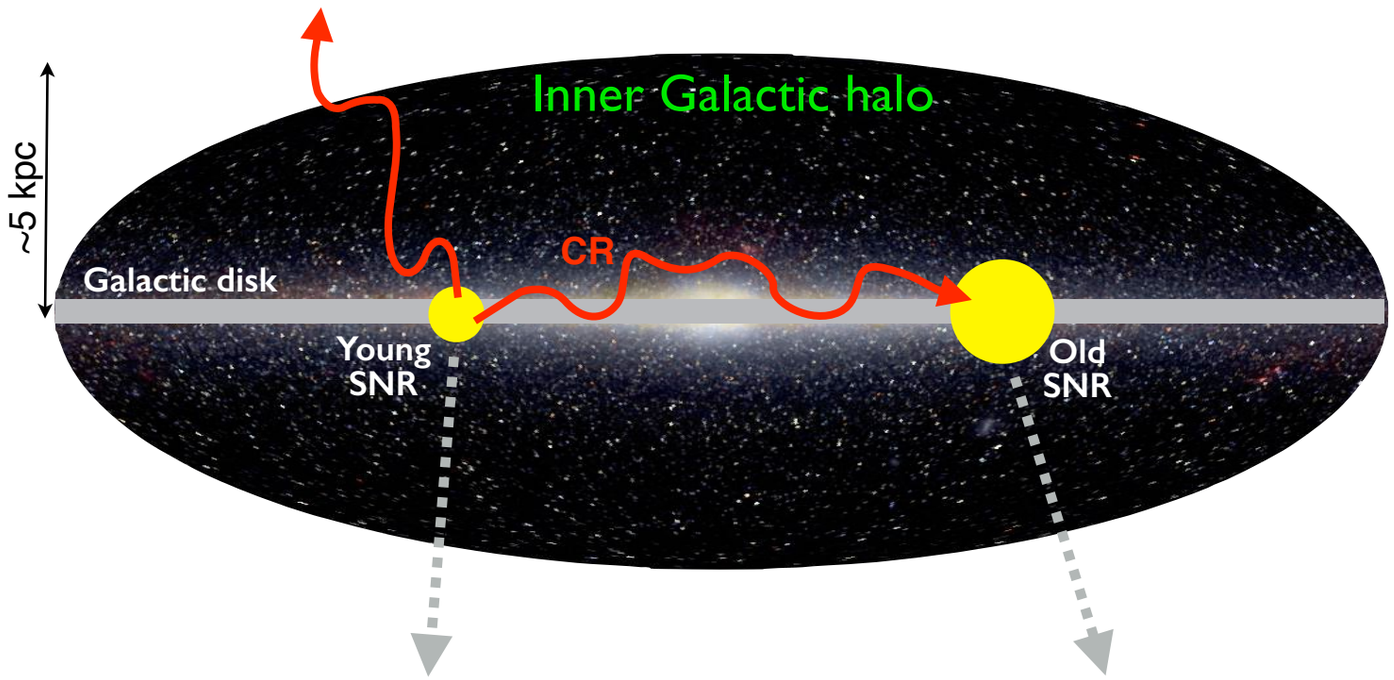
WR-CRs

Both components undergo Galactic propagation including re-acceleration

Thoudam+ 2016, A&A, 595, A33

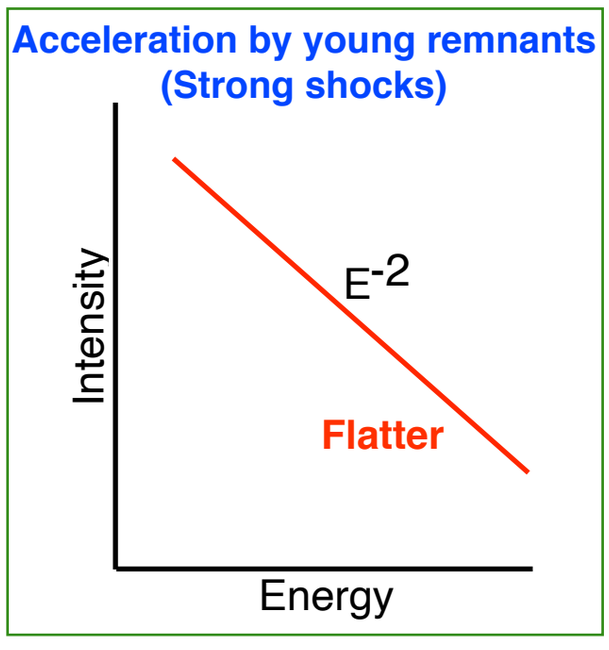
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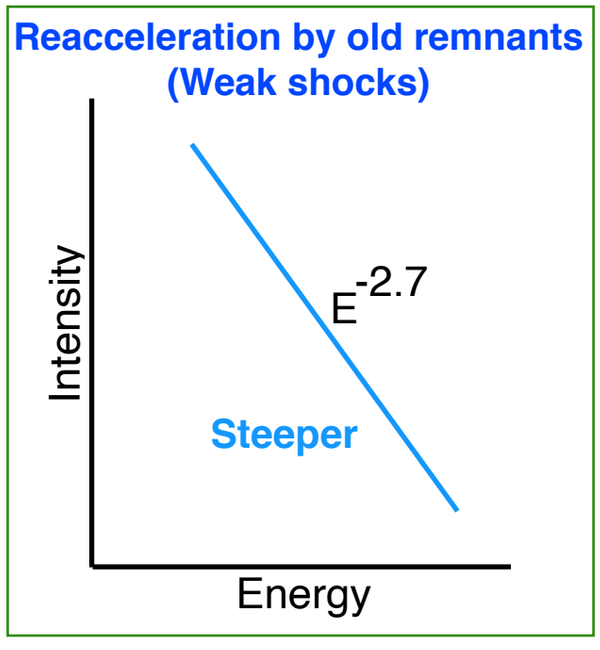
Cosmic-ray propagation including re-acceleration by weak shocks in the Galaxy

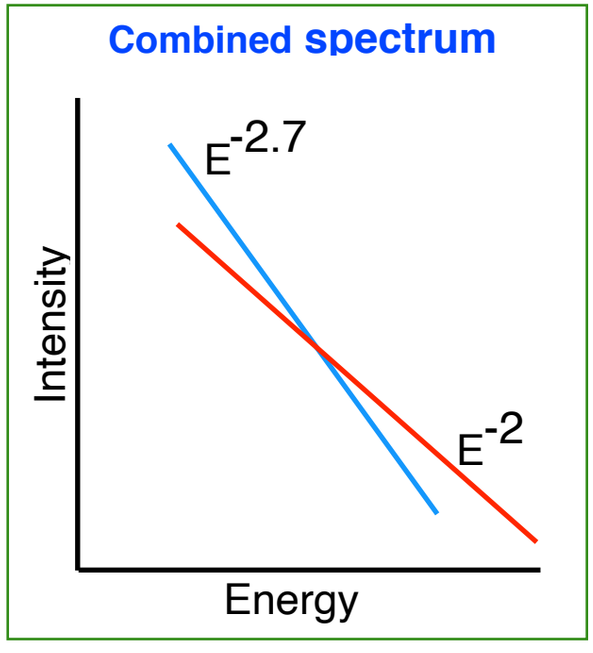


Main assumptions:

- Regular supernova remnants are the main sources of Galactic cosmic rays. **SNR-CRs**
- They are all located in the Galactic disk.
- They produce power-law spectrum of CRs with an exponential cut-off.
- Maximum energy scales with the charge of nuclei.

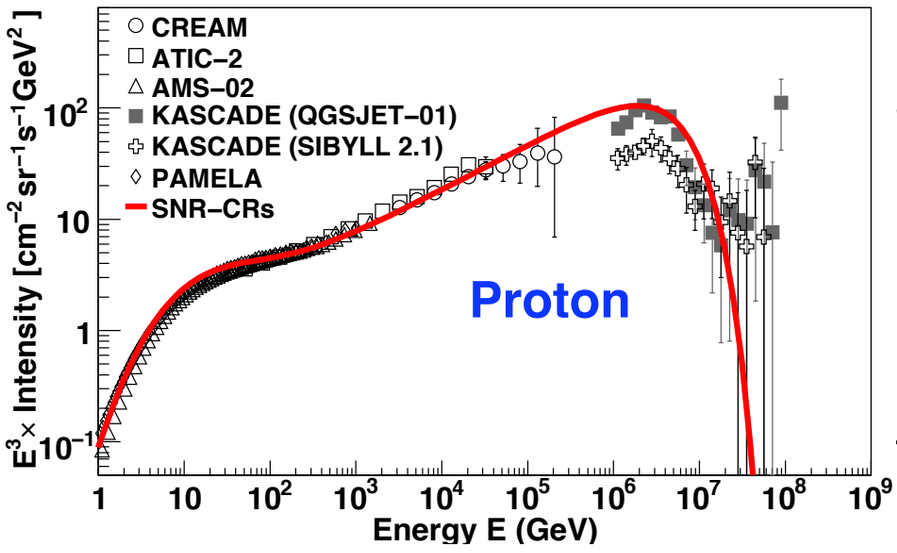


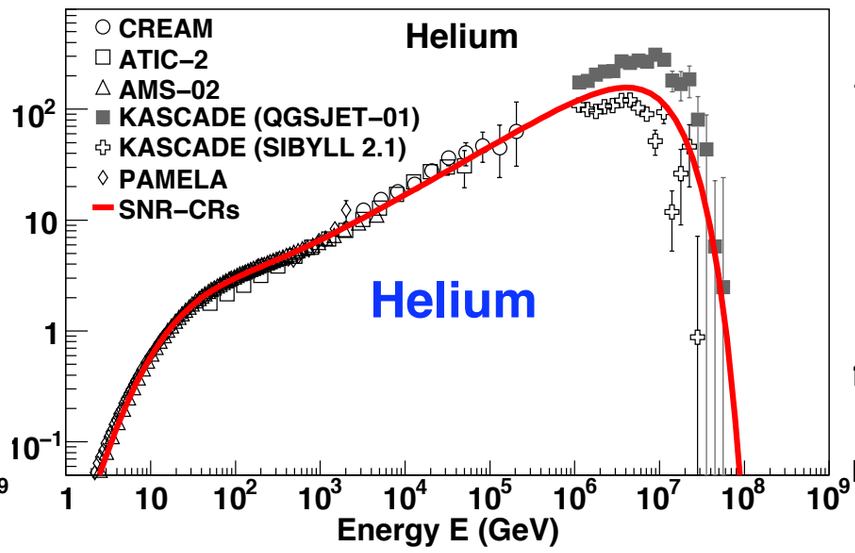


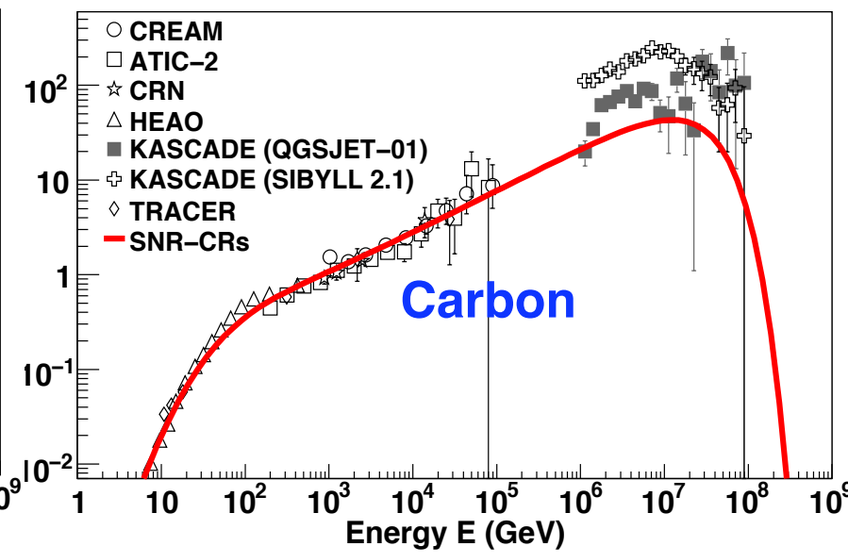


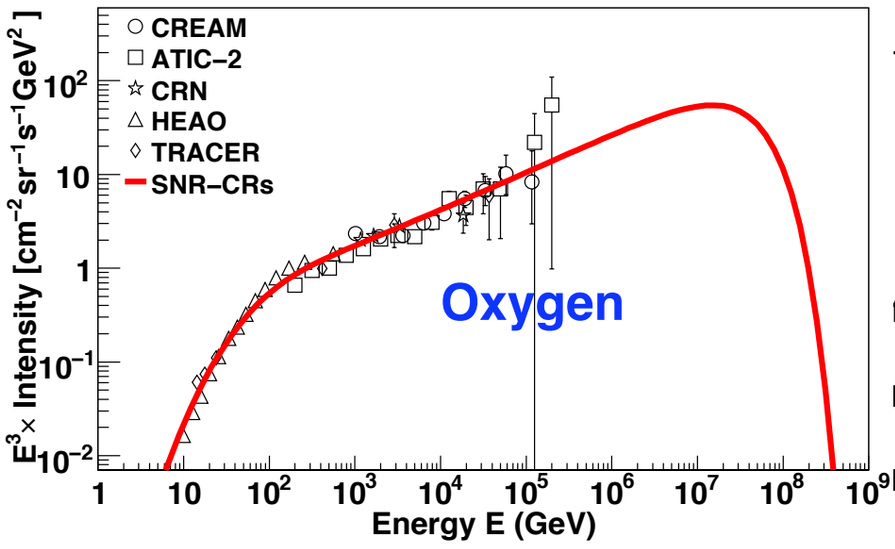
Thoudam & Hörandel 2014, A&A, 567, A33

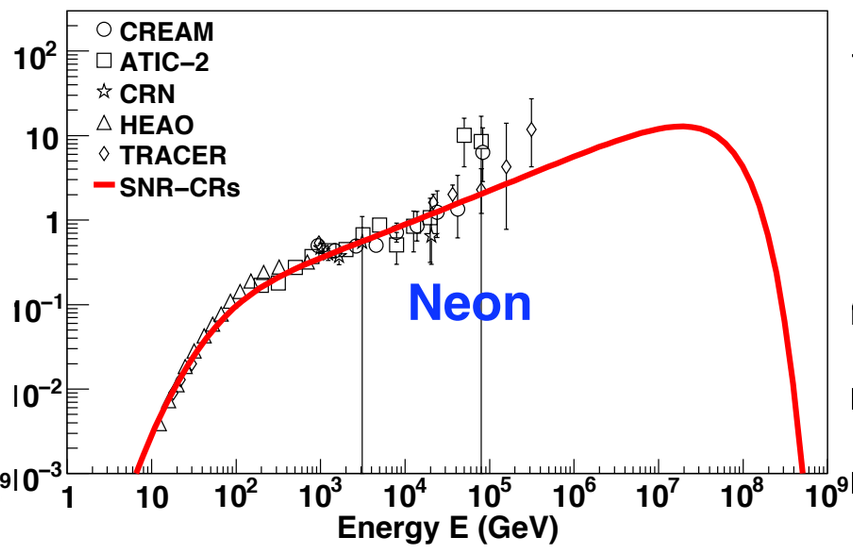
Elemental spectra of SNR-CRs after propagation in the Galaxy

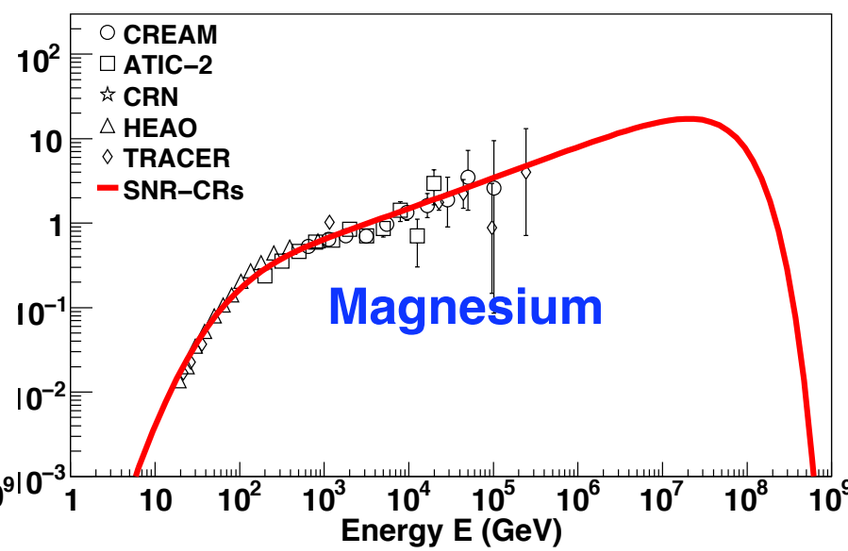


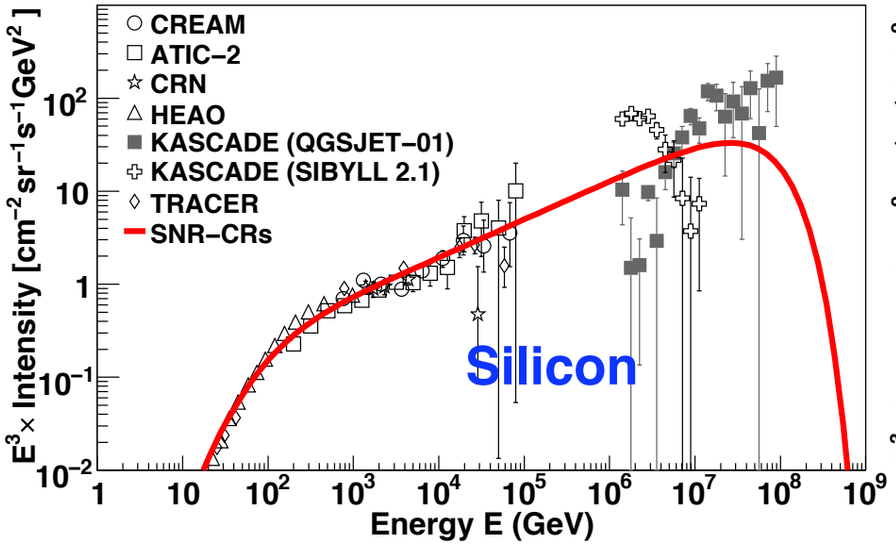


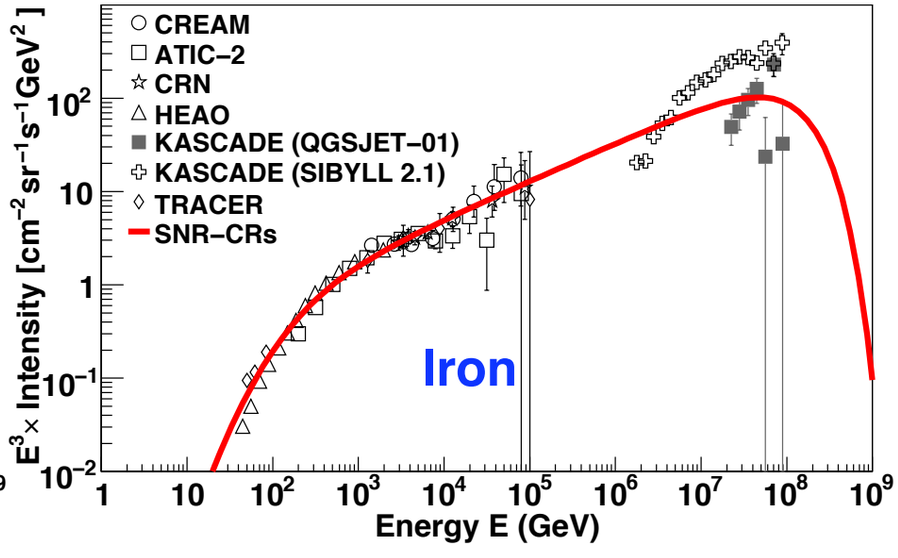








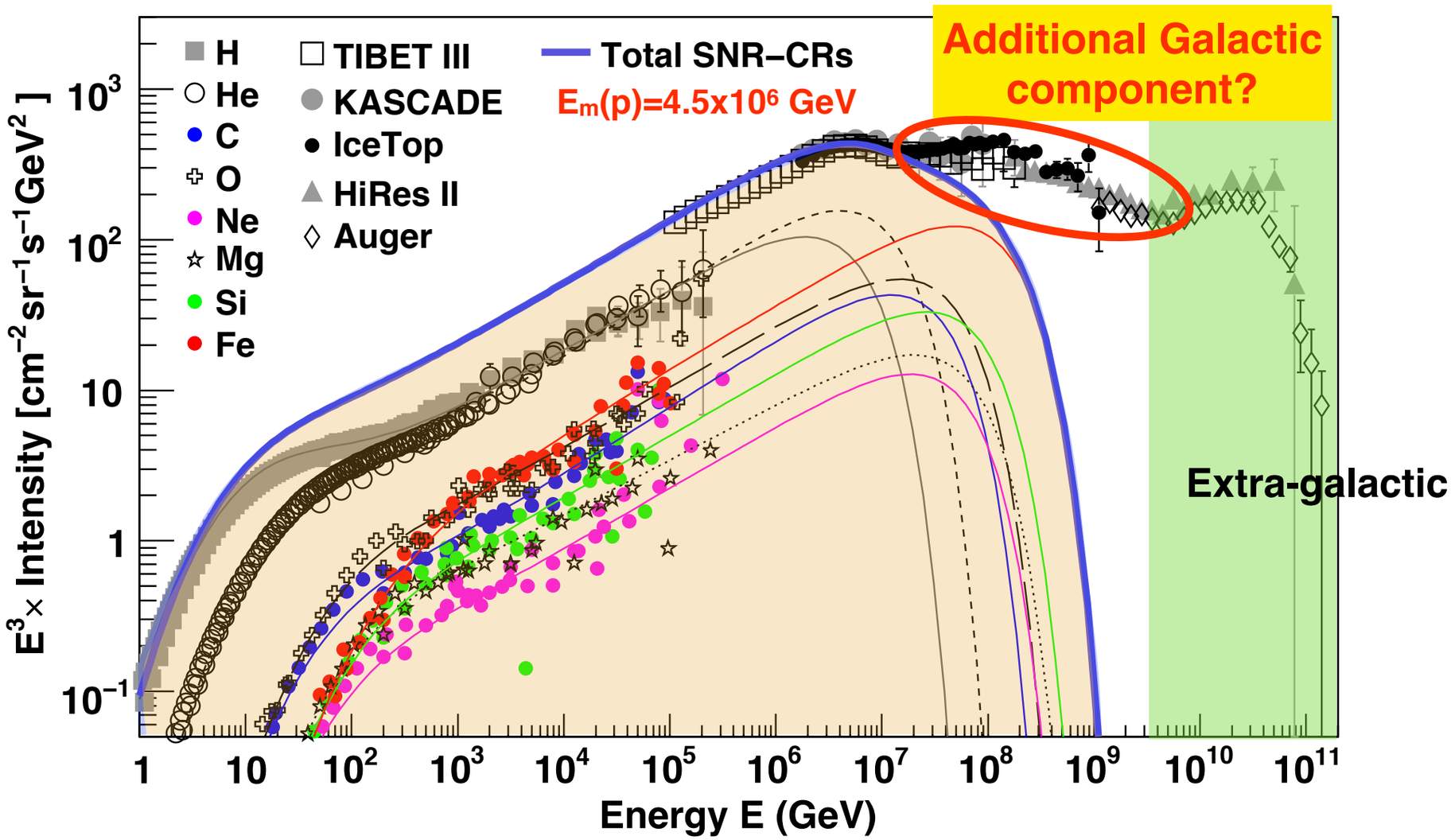




Nuclei	Source index	f(x10^{49} ergs)
Proton	2.24	6.95
Helium	2.21	0.79
Carbon	2.21	2.42×10^{-2}
Oxygen	2.25	2.52×10^{-2}
Neon	2.25	3.78×10^{-3}
Magnesium	2.29	5.17×10^{-3}
Silicon	2.25	5.01×10^{-3}
Iron	2.27	4.95×10^{-3}

(IV) Dominant helium fraction in cosmic rays at ~ $10^{17\text{-}17.5}$ eV

Total contribution of SNR-CRs to the all-particle spectrum



- **The "knee" is dominated by helium nuclei, not by proton.**
- Maximum energy: E_m (Fe)= 26 E_m(p) =1.17x10^8 GeV.
- Prediction close to the "second knee", but not enough in intensity.
- **SNR-CRs alone cannot account for CRs above ~2x10^7 GeV. They contribute only ~30% at 10^8 GeV.**

Thoudam+ 2016, A&A, 595, A33

(IV) Dominant helium fraction in cosmic rays at ~ $10^{17\text{-}17.5}$ eV

Additional Galactic component

We consider two possibilities:

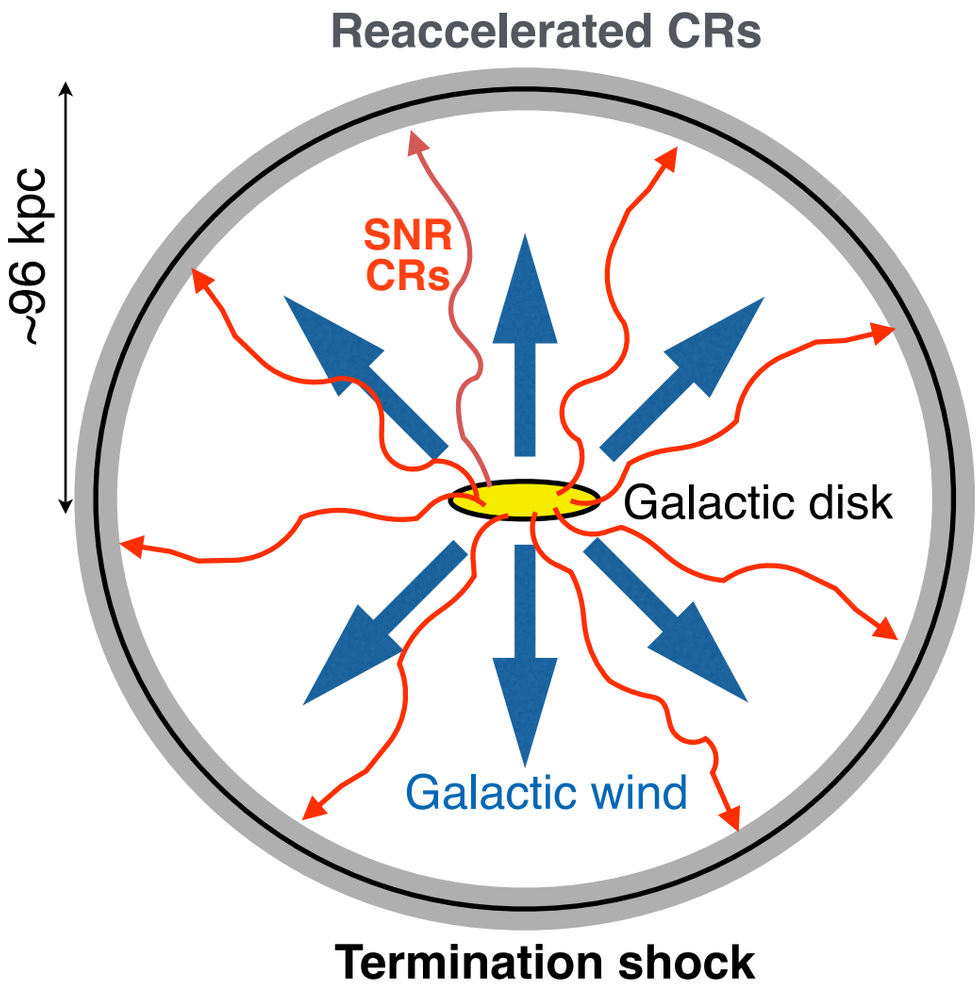
(1) Reacceleration of SNR-CRs escaped into the Galactic halo by Galactic wind termination shocks *(Jokipii 1987, Zirakashvili & Volk 2006)*. **GW-CRs**

(2) Cosmic rays from Wolf-Rayet star supernova explosions in the Galaxy *(Biermann & Cassinelli 1993)*. **WR-CRs**

Thoudam+ 2016, A&A, 595, A33

(IV) Dominant helium fraction in cosmic rays at ~ $10^{17\text{-}17.5}$ eV

Additional Galactic component: Reacceleration by Galactic wind termination shocks (GW-CRs)



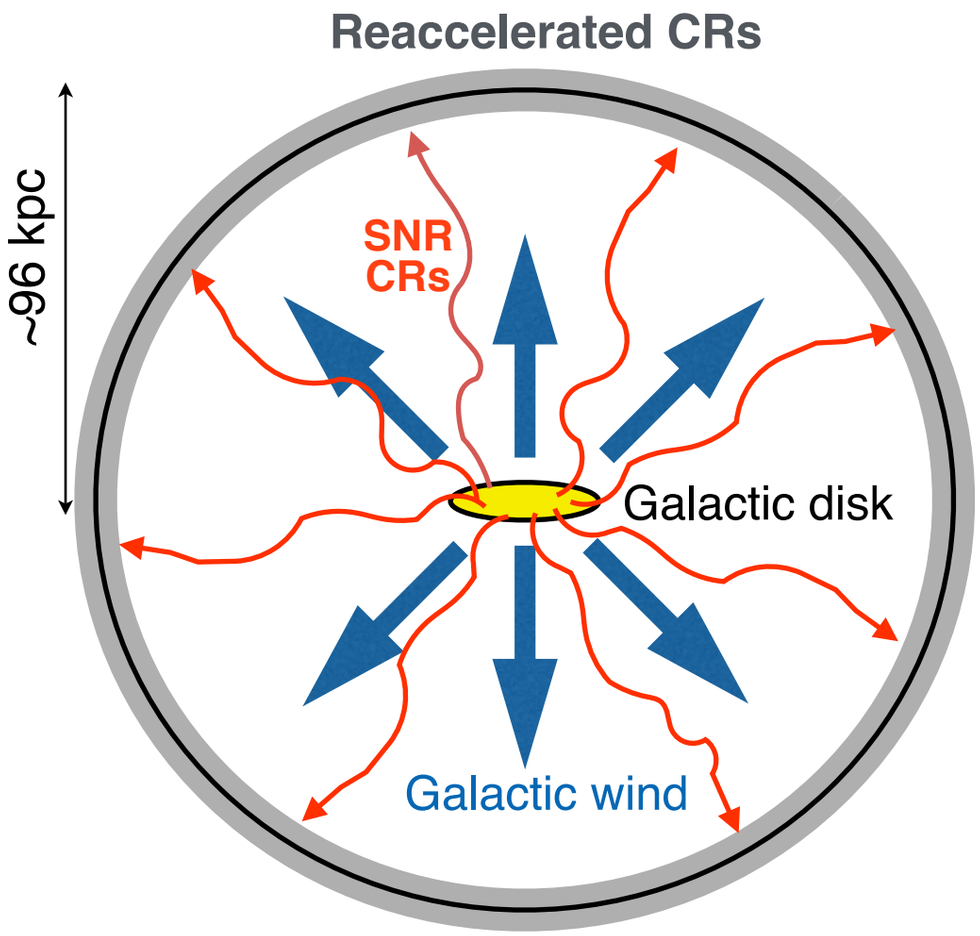
$$\nabla.(D_{\rm w}\nabla N_{\rm w} - \boldsymbol{V}N_{\rm w}) + \frac{\partial}{\partial p}\left\{\frac{\nabla.\boldsymbol{V}}{3}pN_{\rm w}\right\} = -Q_{\rm esc}\delta(\boldsymbol{r})$$

Diffusion	Convection	Adiabatic loss	Escape rate of SNR-CRs

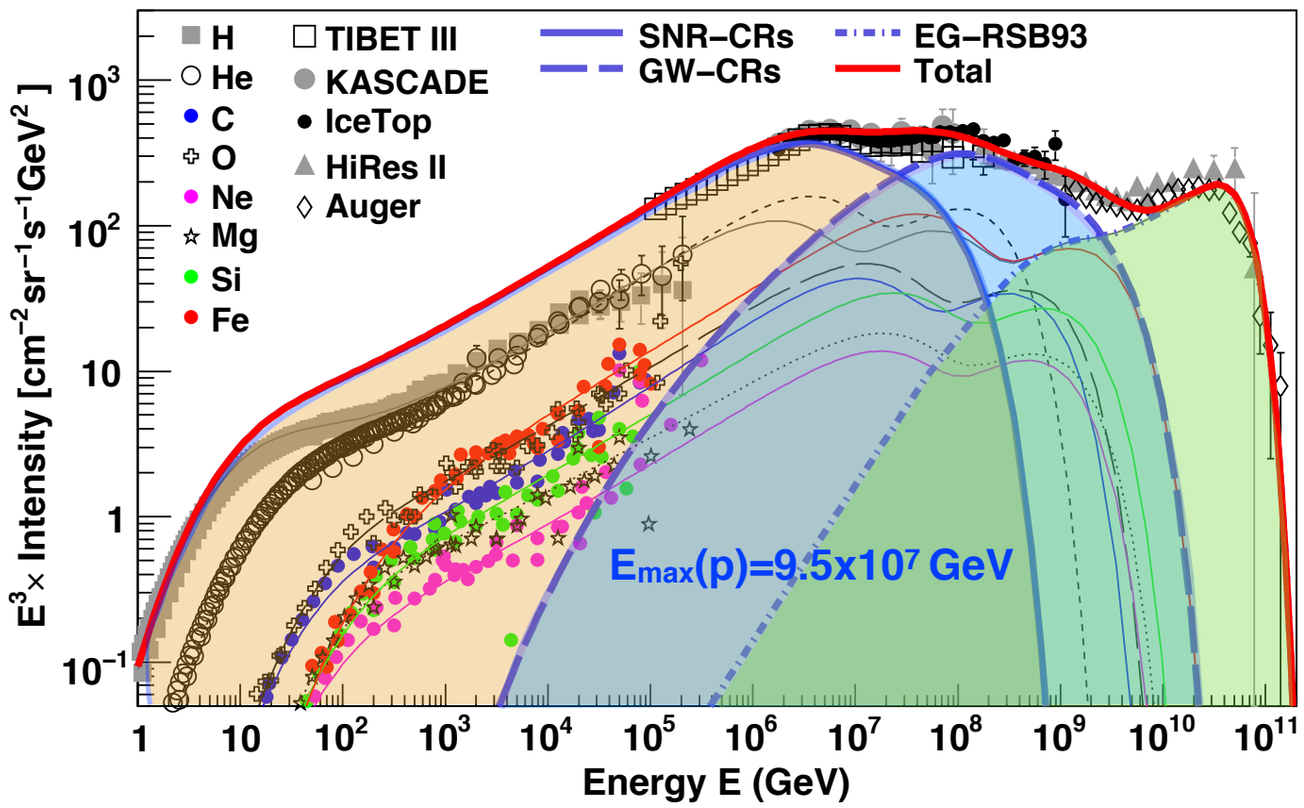
Thoudam+ 2016, A&A, 595, A33

(IV) Dominant helium fraction in cosmic rays at ~ $10^{17\text{-}17.5}$ eV

Additional Galactic component: Reacceleration by Galactic wind termination shocks (GW-CRs)



Thoudam+ 2016, A&A, 595, A33



$$\nabla.(D_{\rm w}\nabla N_{\rm w} - \boldsymbol{V}N_{\rm w}) + \frac{\partial}{\partial p}\left\{\frac{\nabla.\boldsymbol{V}}{3}pN_{\rm w}\right\} = -Q_{\rm esc}\delta(\boldsymbol{r})$$

Diffusion | Convection | Adiabatic loss | Escape rate of SNR-CRs

$D_{\rm w}$=10 $D_{\rm gal}$

$V=V_0 r$, V_0=15 km/s/kpc

$f_{\rm inj}$ =14.5%

$E_{max}(p)=9.5x10^7$ GeV

- **Extragalactic cosmic-ray model:**
  - *Rachen, Stanev, & Biermann, 1993, A&A, 273, 377*
  - All-proton with E^{-2} source spectrum and exponential cut-off at 10^{11} GeV

Thoudam+ 2016, A&A, 595, A33

(IV) Dominant helium fraction in cosmic rays at ~ $10^{17\text{-}17.5}$ eV

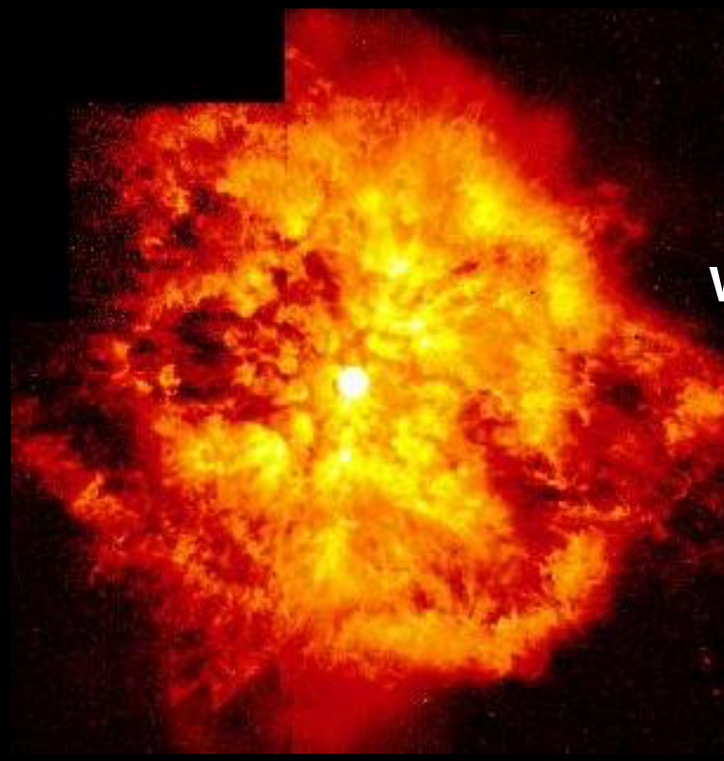
Additional Galactic component: CRs from Wolf-Rayet star supernova explosions (WR-CRs)

- **Mostly located in the Galactic disk like the regular supernova remnants.**
- **Rare, massive stars with fast winds; mass~ 20 M_s and frequency ~1/210 yr^{-1} = 1/7 SNe**
- **Lack of hydrogen in the wind.**
- **Short-lived before exploding into supernova.**
- **High magnetic field, ~ 100 G or even more** (Chevrotiere+ 2013, 2014).
- **Accelerate particles up to ~ 10^{18} eV** (Biermann & Cassinelli 1993, Stanev+ 1993).

Wolf-Rayet wind composition

Nuclei	C/He=0.1	C/He=0.4
Proton	0	0
Helium	1.0	1.0
Carbon	0.1	0.4
Oxygen	3.19×10^{-2}	7.18×10^{-2}
Neon	0.42×10^{-2}	1.03×10^{-2}
Magnesium	2.63×10^{-4}	6.54×10^{-4}
Silicon	2.34×10^{-4}	5.85×10^{-4}
Iron	0.68×10^{-4}	1.69×10^{-4}

Pollock+ 2005



WR124

Credit: NASA, HST

Thoudam+ 2016, A&A, 595, A33

(IV) Dominant helium fraction in cosmic rays at ~ $10^{17\text{-}17.5}$ eV

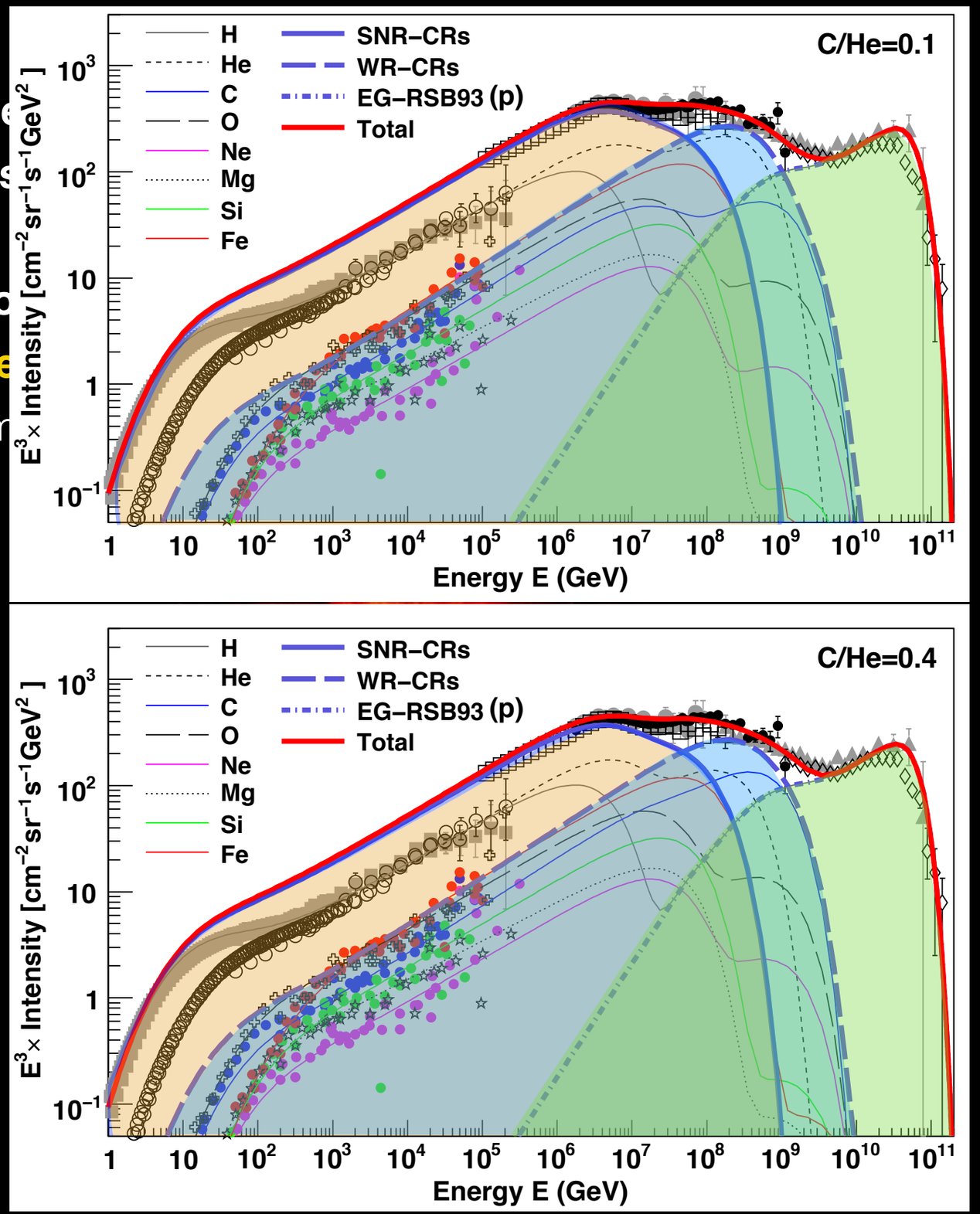
Additional Galactic component: CRs from Wolf-Rayet star supernova explosions (WR-CRs)

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- **Lack of hydrogen in the wind.**
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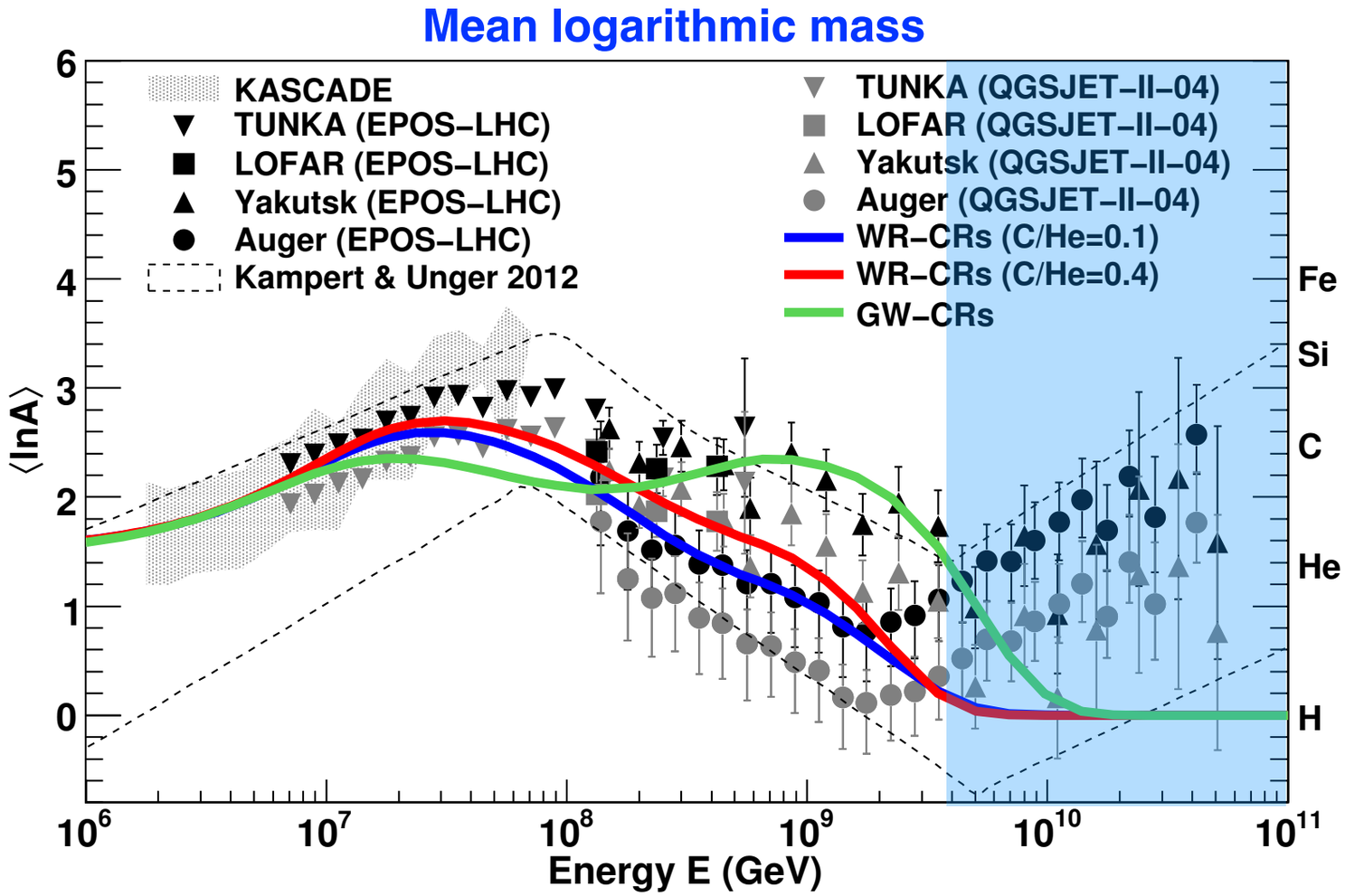
Pollock+ 2005



Thoudam+ 2016, A&A, 595, A33

(IV) Dominant helium fraction in cosmic rays at ~ $10^{17\text{-}17.5}$ eV

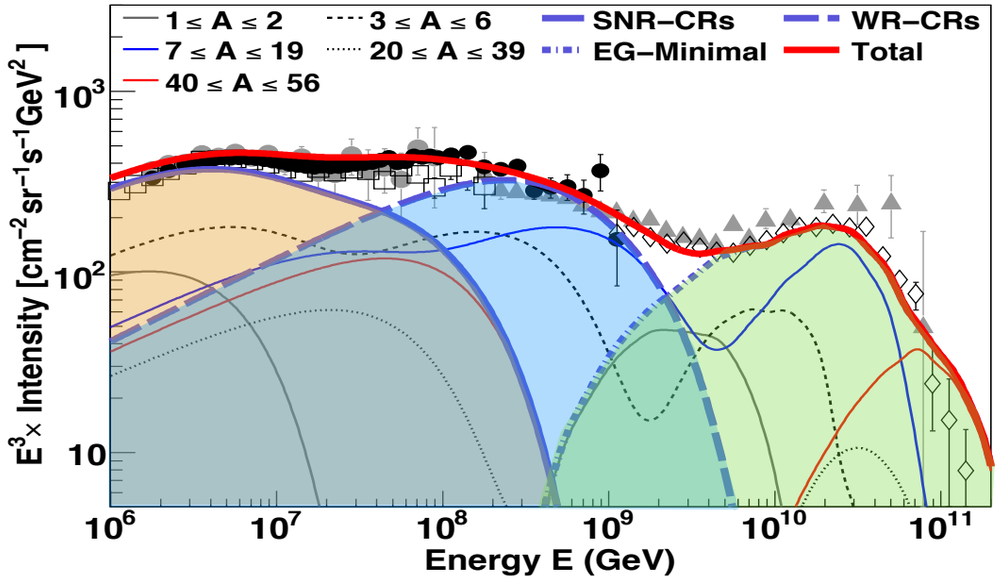
Cosmic-ray composition predicted by the different scenarios



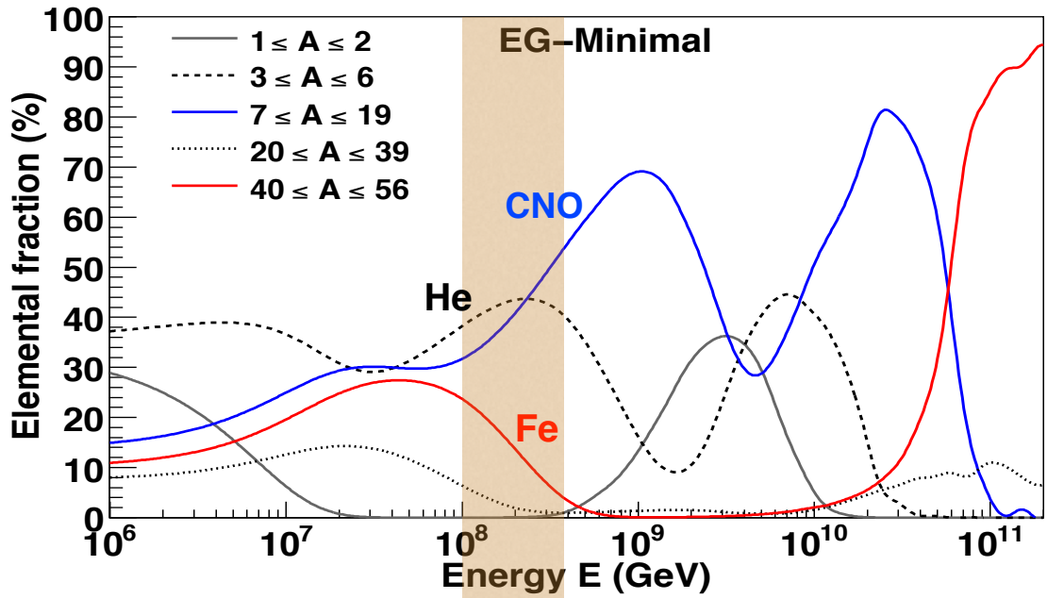
- Galactic wind scenario does not fit well the observed composition.
- Both the Wolf-Rayet scenarios explain the data quite well.
- Bad fit above the Ankle is due to the all-proton extra-galactic model
  (Rachen, Stanev, & Biermann, 1993, A&A, 273, 377)

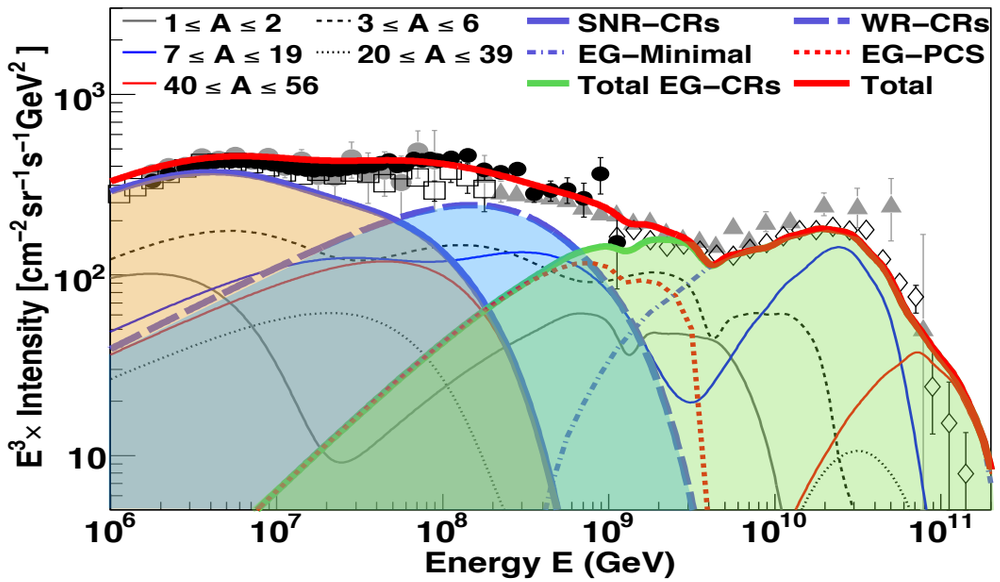
Thoudam+ 2016, A&A, 595, A33

Tests of Wolf-Rayet (C/He=0.4) case with different extra-galactic models

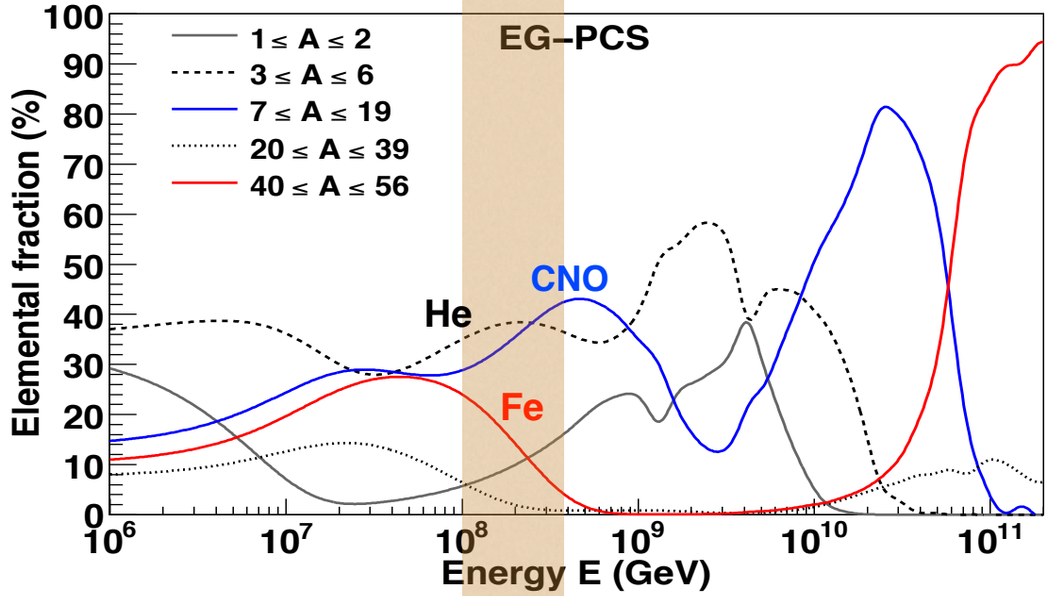


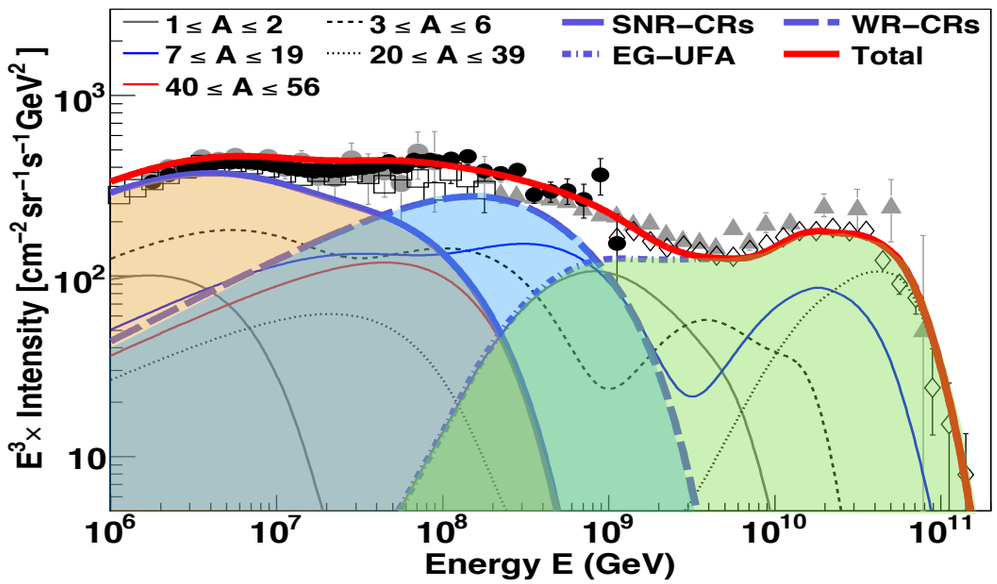
EG-Minimal
(di Matteo+ 2015)



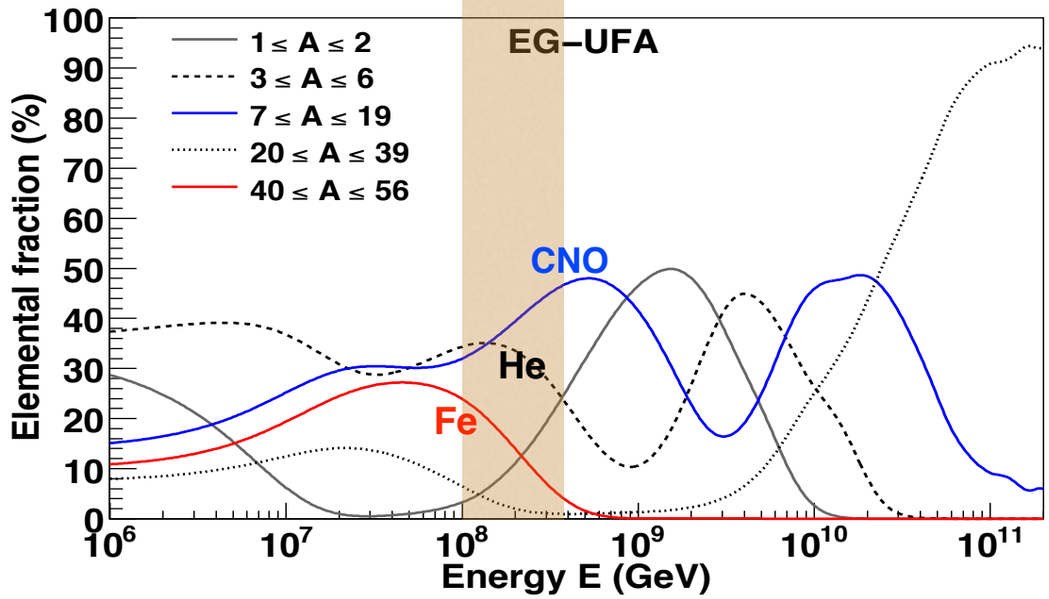


EG-PCS
(Rachen 2016)



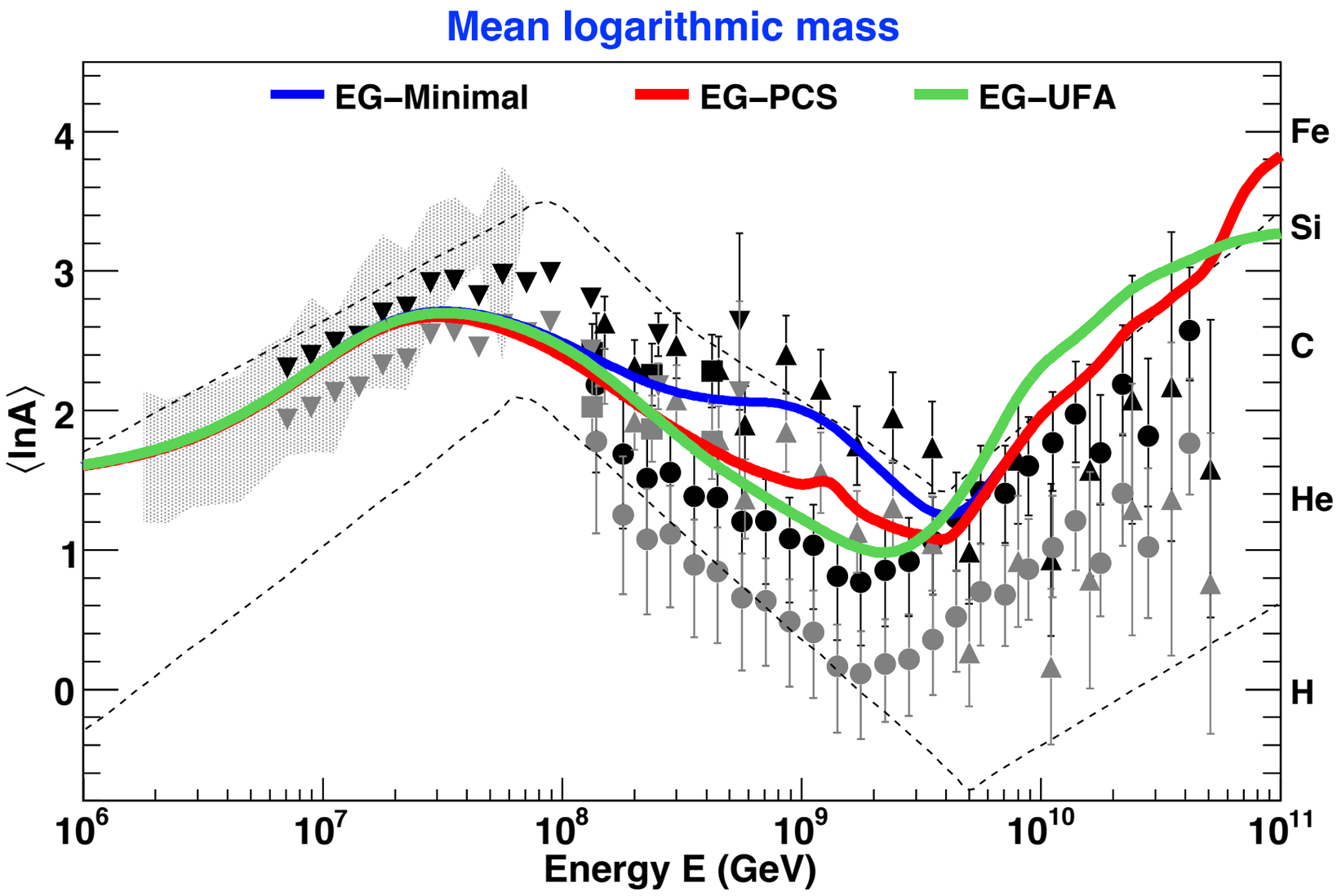


EG-UFA
(Unger+ 2016)



(IV) Dominant helium fraction in cosmic rays at ~ $10^{17\text{-}17.5}$ eV

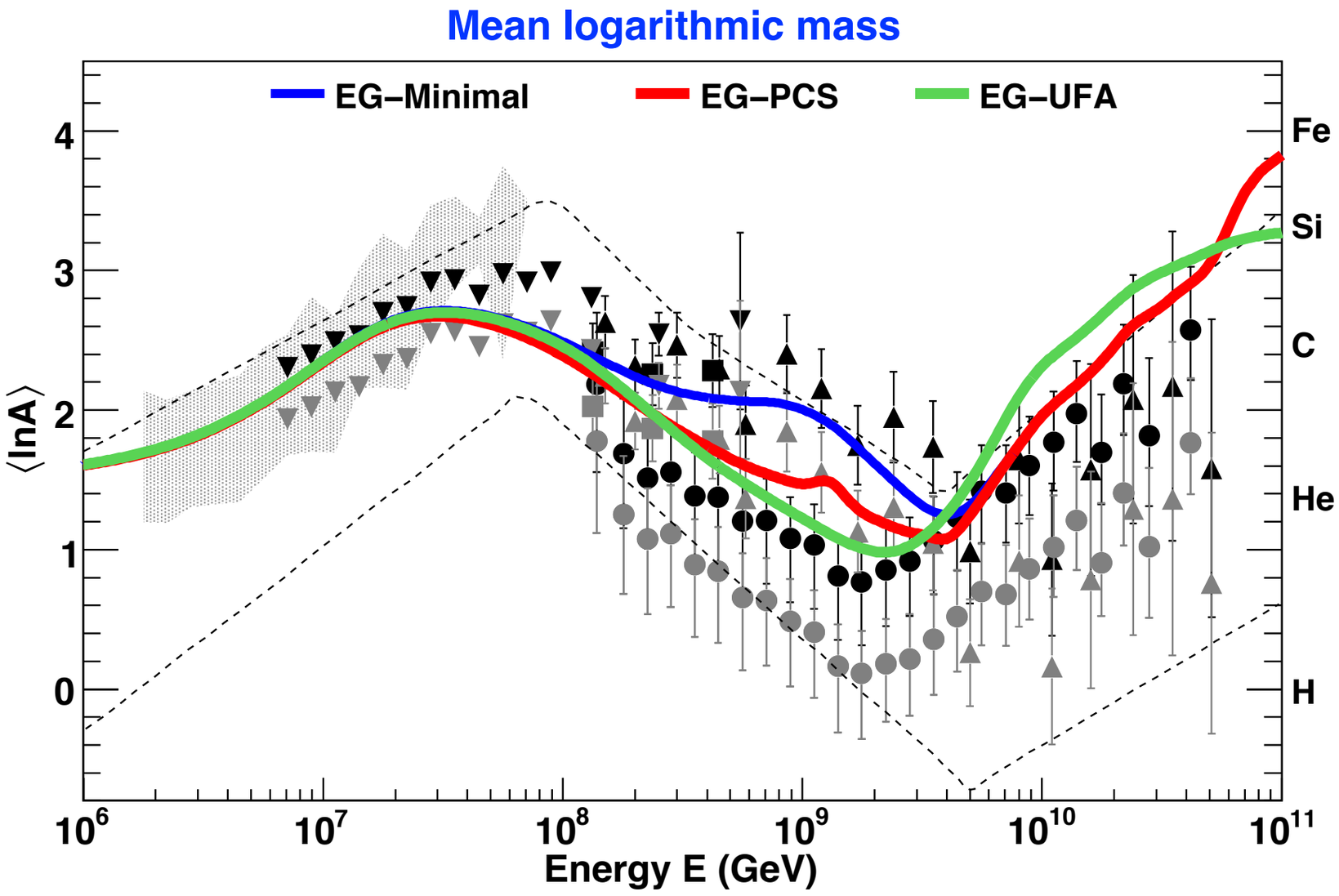
Tests of Wolf-Rayet (C/He=0.4) case with different extra-galactic models: CR Composition



Thoudam+ 2016, A&A, 595, A33

(IV) Dominant helium fraction in cosmic rays at ~ $10^{17\text{-}17.5}$ eV

Tests of Wolf-Rayet (C/He=0.4) case with different extra-galactic models: CR Composition



Main conclusion

- Wolf-Rayet scenario reproduces the observed light nuclei at ~$10^{17\text{-}17.5}$ eV.
- The result is quite robust to the model of the extra-galactic cosmic rays.

Thoudam+ 2016, A&A, 595, A33

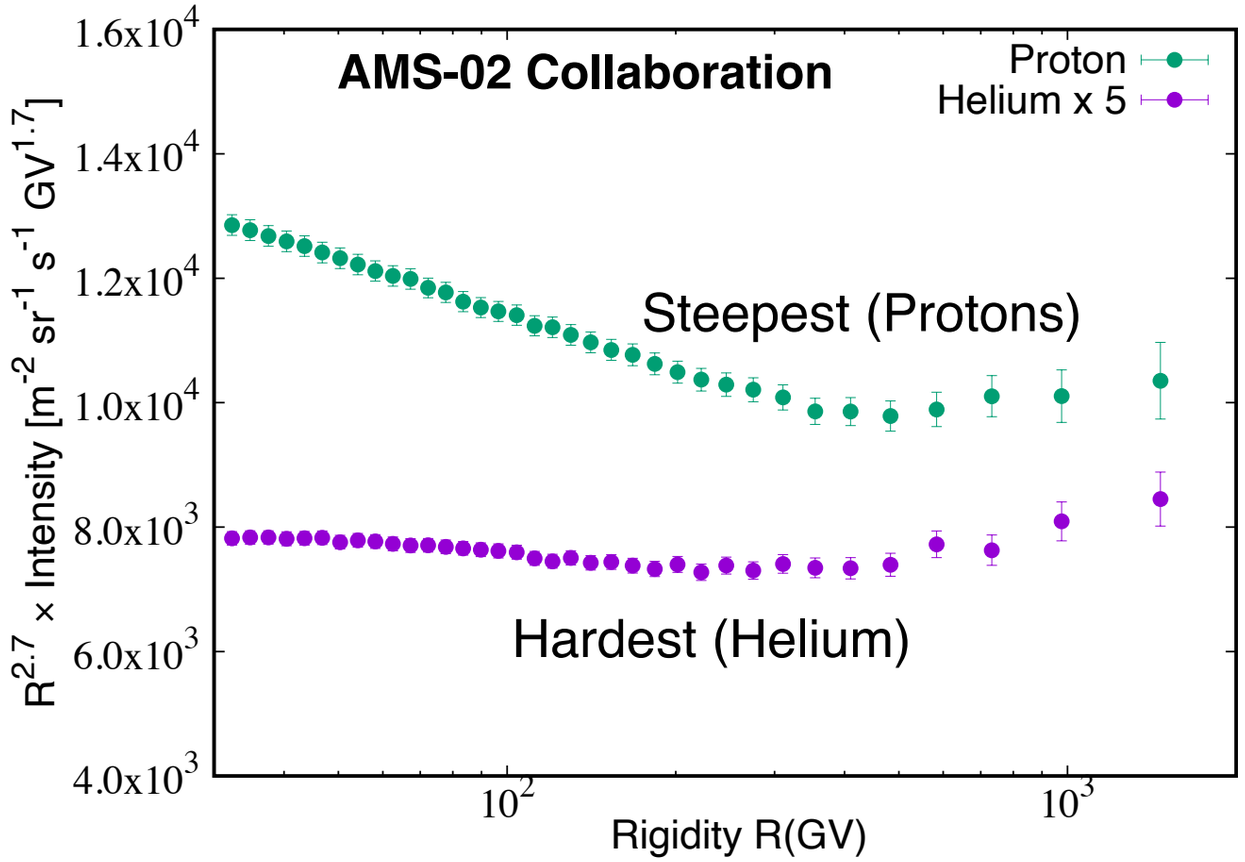
(V) Different spectral shapes between primary CRs below 10^{14} eV

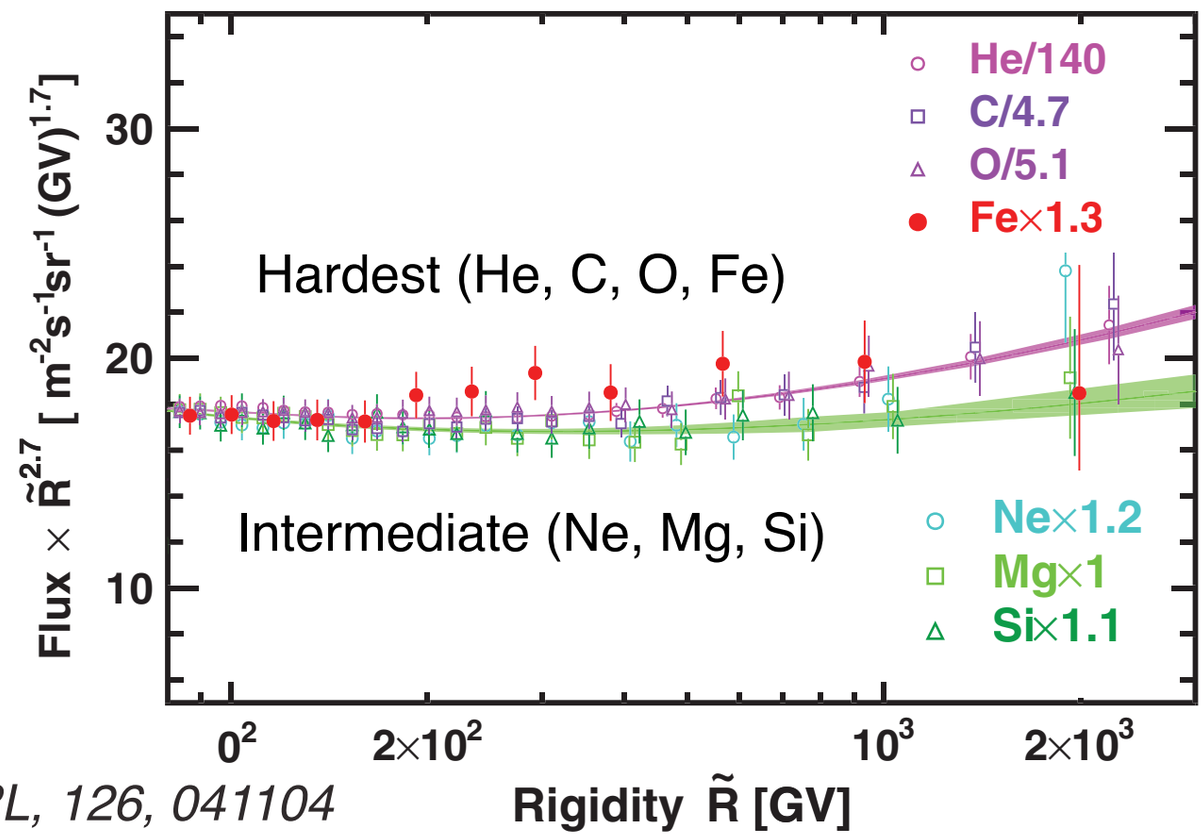
2015, PRL, 114,171103
2015, PRL, 115, 211101

- **AMS-02, CALET, CREAM, DAMPE,…..**
- **Different spectral shapes between primary cosmic-ray species.**
- **Protons: Steepest spectrum.**
- **Then, Ne, Mg, Si**
- **He, C, O, Fe: Hardest.**

Diffusive shock acceleration theory + propagation: Spectral index independent of the type of nuclei.

Krymsky 1977, Bell 1978
Blandford & Eichler 1987
Ginzburg & Ptuskin 1976





2021, PRL, 126, 041104

(V) Different spectral shapes between primary CRs below 10^{14} eV

Model

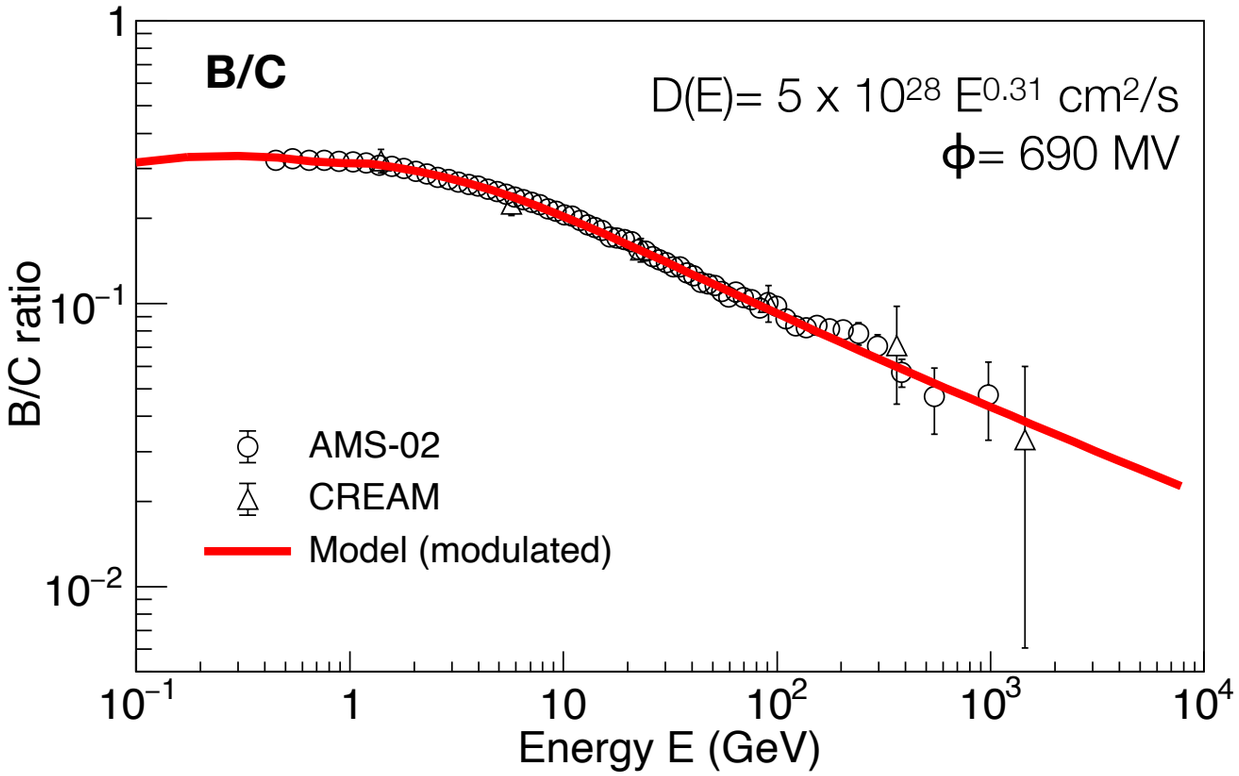
***The same two-component model of Galactic cosmic rays described earlier:**

- 1st component: CRs from regular supernovae
- 2nd component: CRs from Wolf-Rayet (WR) supernovae
- Both components undergo Galactic propagation including re-acceleration

***Here, we allow the source indices of the two components to vary. In the previous work, they were kept the same.**

(V) Different spectral shapes between primary CRs below 10^{14} eV

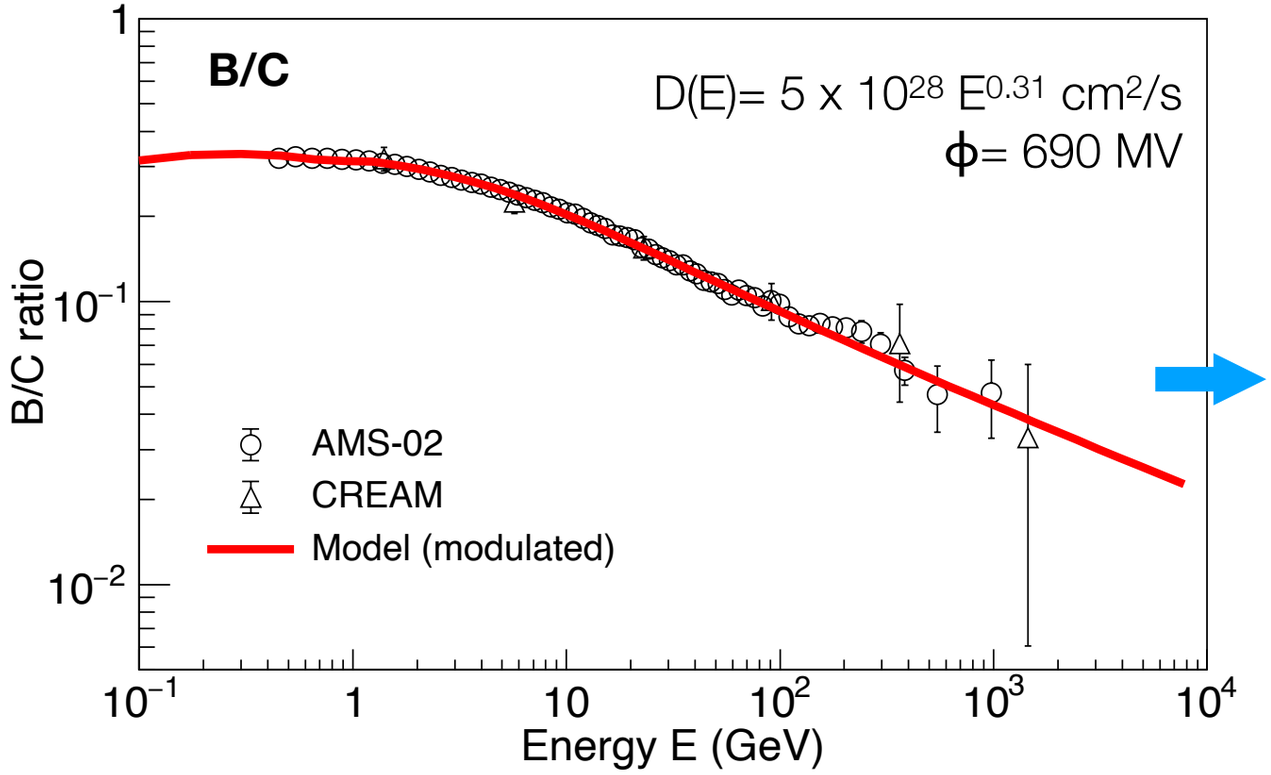
Results: B/C and CR primary spectra (fit to the AMS data only)

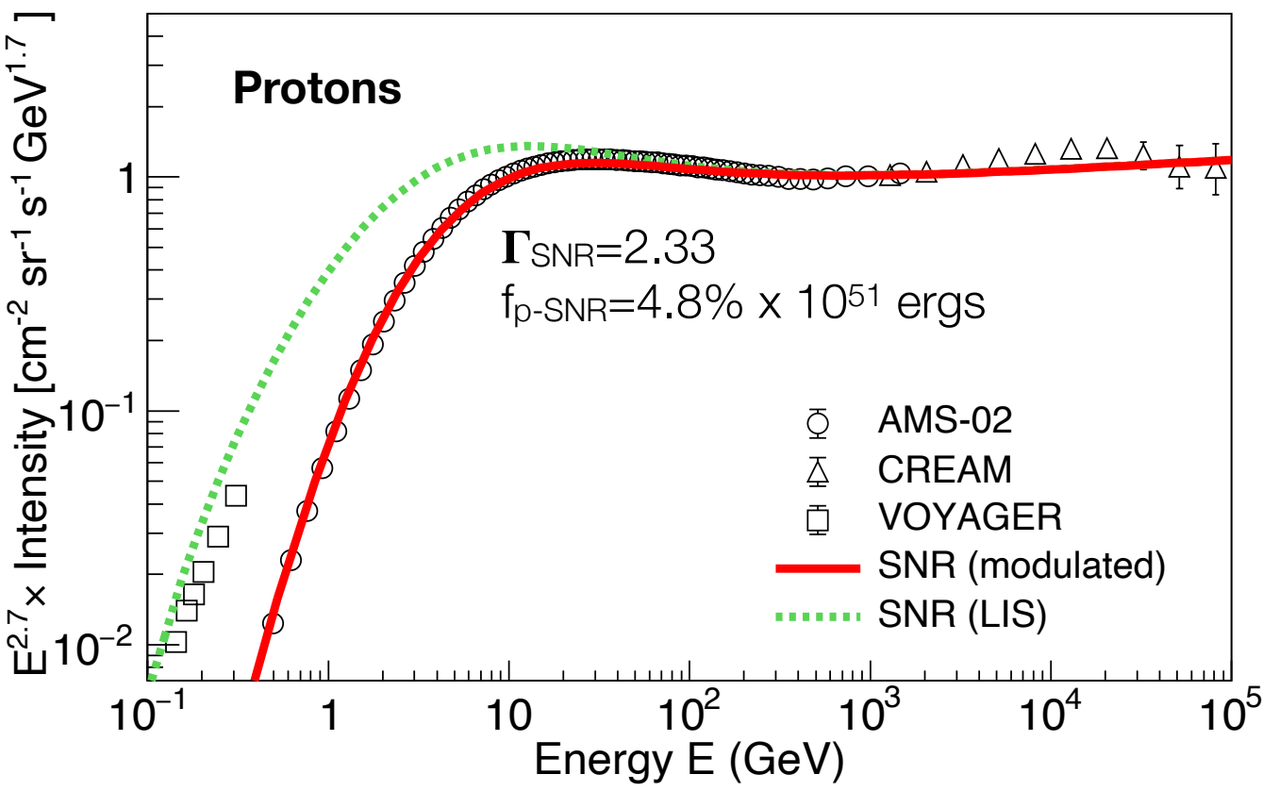


(1) B/C fit fixes the diffusion coefficient.

(V) Different spectral shapes between primary CRs below 10^{14} eV

Results: B/C and CR primary spectra (fit to the AMS data only)



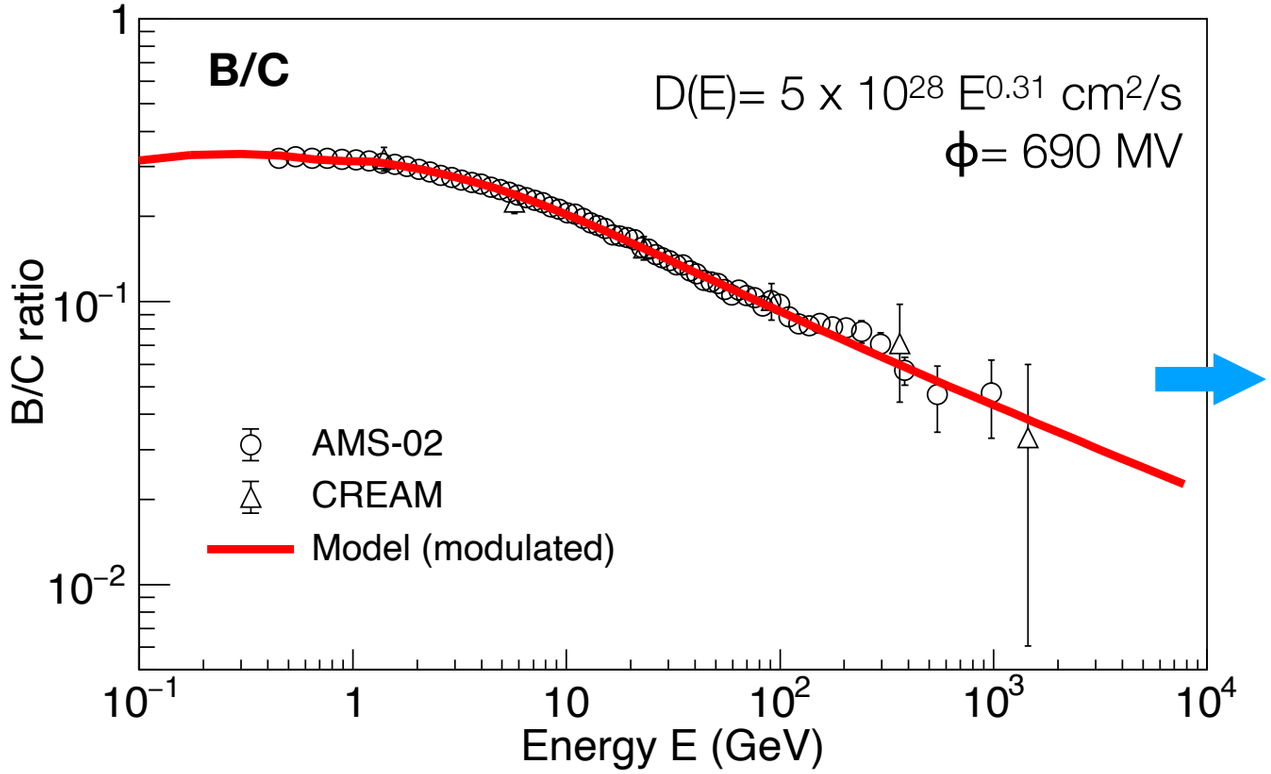


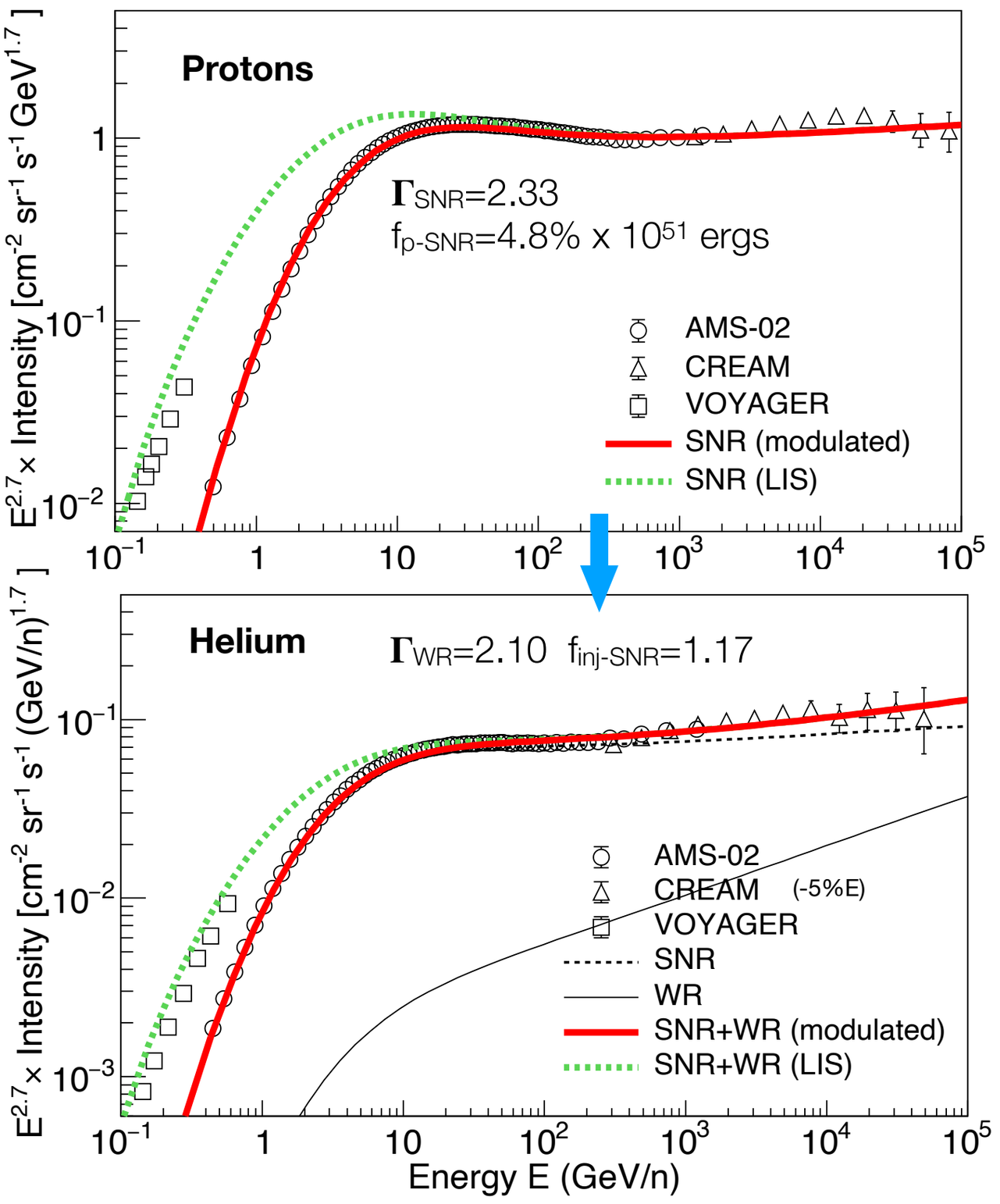
(1) B/C fit fixes the diffusion coefficient.

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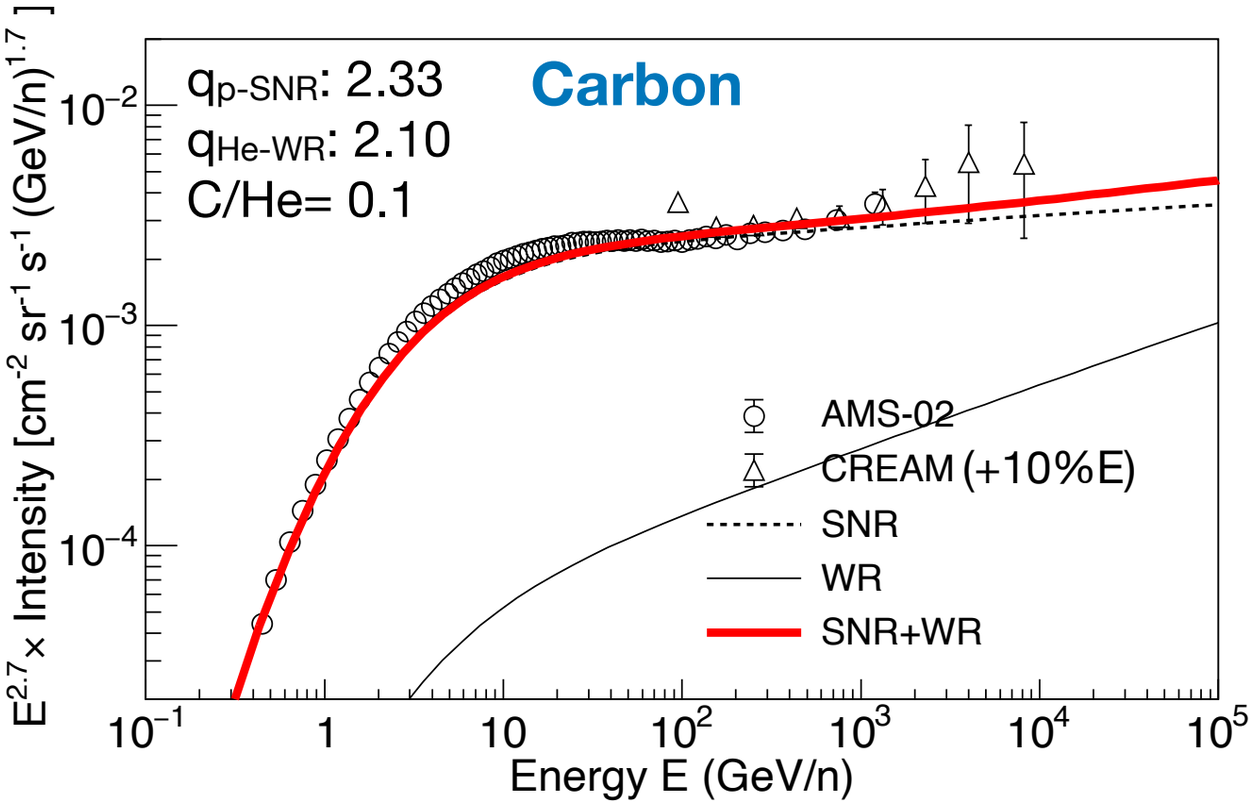
(1) B/C fit fixes the diffusion coefficient.

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(3) Helium spectrum fit fixes the source index of the WR-SNRs.

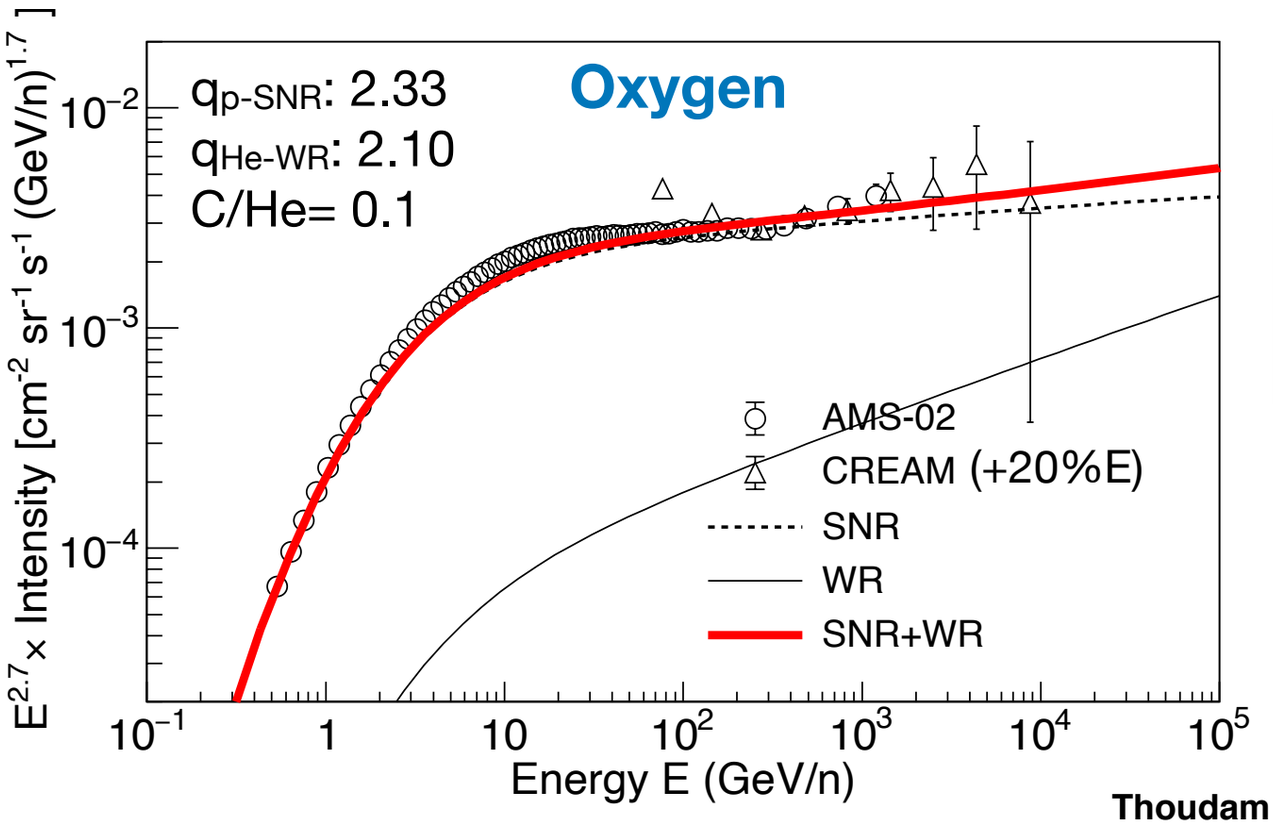
(V) Different spectral shapes between primary CRs below 10^{14} eV

Results: B/C and CR primary spectra (fit to the AMS data only)



$f_{C\text{-}SNR}$: 0.018% x 10^{51} ergs
$f_{inj\text{-}C\text{-}SNR}$: 13.0
$f_{C\text{-}WR}$: 0.008% x 10^{51} ergs
$f_{inj\text{-}C\text{-}WR}$: 0.28 (w.r.t He)

D(E) and source index remain the same as for the protons and helium.



$f_{O\text{-}SNR}$: 0.02% x 10^{51} ergs
$f_{inj\text{-}O\text{-}SNR}$: 8.0
$f_{O\text{-}WR}$: 0.01% x 10^{51} ergs
$f_{inj\text{-}O\text{-}WR}$: 1.2 (w.r.t He)

Thoudam + 2021, 37th International Cosmic-ray Conference, Berlin (Germany)

Next, we will learn about the measurements of cosmic particles.

Thoudam & Hörandel, 2012, MNRAS, 419, 624